X'mas SnowGlobe TableTop Toy for micro:bit

BBC micro:bit v2

- ***User Manual*** -

(5 ~ 8 Hours)

https://.../MicroBit_V2_Edition

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- Content -

- **Part 1** - Introduction
  - Why SnowGlobe?
  - Pre-requisite for the guided tour
- **Part 2** - X'MAS SnowGlobe Construction
  - SnowGlobe Peripheral Board
  - micro:bit Carrier Board
  - SnowGlobe Power Options
- **Part 3** - Carrier Board programming (micro:bit)
  - "Hello World" of embedded world - Blink
  - "Let there be Light" - Detecting ambient light intensity
  - Interaction - Detecting Touch
  - "Hark the herald angels sing" - Playing a melody
  - Controlling the Neopixels (RGB LEDs)
  - Knowing & Showing Time - OLED & RTC
- **Part 4** - Project 1: SnowGlobe Clock with & Automatic Ambient Nite-light demonstration
  - SnowGlobe Ambient + Nite-light functional description.
  - Loading the working code.
  - Student Challenge

Santa & His Elf

- Part 1 -

Introduction

Why SnowGlobe?

Because every learner needs a moment of wonder. A spark that turns curiosity into the courage to create🔨.

This **Product** was born from a **<u>simple observation</u>**: when students build something that feels special 🪄, they **stay engaged** longer, **ask deeper questions**, and **take greater pride** in their **learning**.

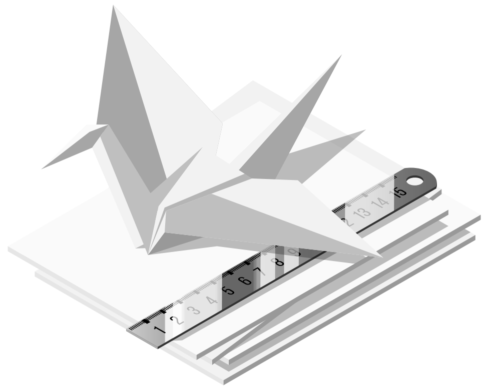
A snow globe captures that feeling **perfectly👍 :**

Shake it, and a quiet world🌏 comes alive. **Build one,** and you create your own tiny universe🪐 of possibilities.

X'mas SnowGlobe TableTop Toy takes that magic🪄 and turns it into a hands-on STEM story. Students aren't just assembling parts; they're ultimately crafting a piece of art powered by their own code - They see how a line of logic becomes a combination of colorful light, how sensors respond to their touch or environment, and how learning STEM (Coding) can transforms something **DULL** to something **SPECTACULAR**.

The SnowGlobe becomes a **symbol**🌟 of what they can build when **imagination** meets **engineering**.

SnowGlobe isn't just a project! It's an invitation to create something meaningful, memorable, and a little bit magical. Treat it in the end as a keepsake - a reminder of a moment when you, a learner, discovered **"I made this myself."**🎄



Pre-requisite for the Guided-tour 🚌

This project might be unlike anything you have done before. It involves **two** disciplines from the acronym **STEAM**—**T**echnology & **A**rt. Thus in our opinion, it would benefit greatly if you approach this project challenge armed with some prior experience & knowledge.

1. The construction & assembly portion will be a breeze if you have taken apart your toys or done simple electronics in the past. However, if you want to brush up your skill in doing electronics surgery or handling electronics, you can refer to the following link:

Dissect An Old Toy Series by Scitech WA:
Dissect An Old Toy Part 1 : Think
Dissect An Old Toy Part 2 : Build
Dissect An Old Toy Part 3 : Test

2. We will not be starting micro:bit graphical coding from scratch in this user manual. There are many resources online to get you started on your journey with your micro:bit:

Micro:bit Education Foundation
Youtube - Get started with the micro:bit

- Part 2 -

X'MAS SnowGlobe Construction

X’mas SnowGlobe Peripheral-Board

- The SnowGlobe Peripheral Board is the core of the entire build - it’s where all the interaction happens.
  It houses/consists of:

  - **OLED** display for messages, animations, or even a clock output
  - **NeoPixel** LEDs for colorful lighting effects
  - Standard **LEDs** for accent lighting
  - **Light sensor** for ambient-aware brightness
  - **Capacitive-touch input** for user interaction
  - 2 Pieces of Wood, a piece of Acrylic & a bunch of 3D printed trinkets

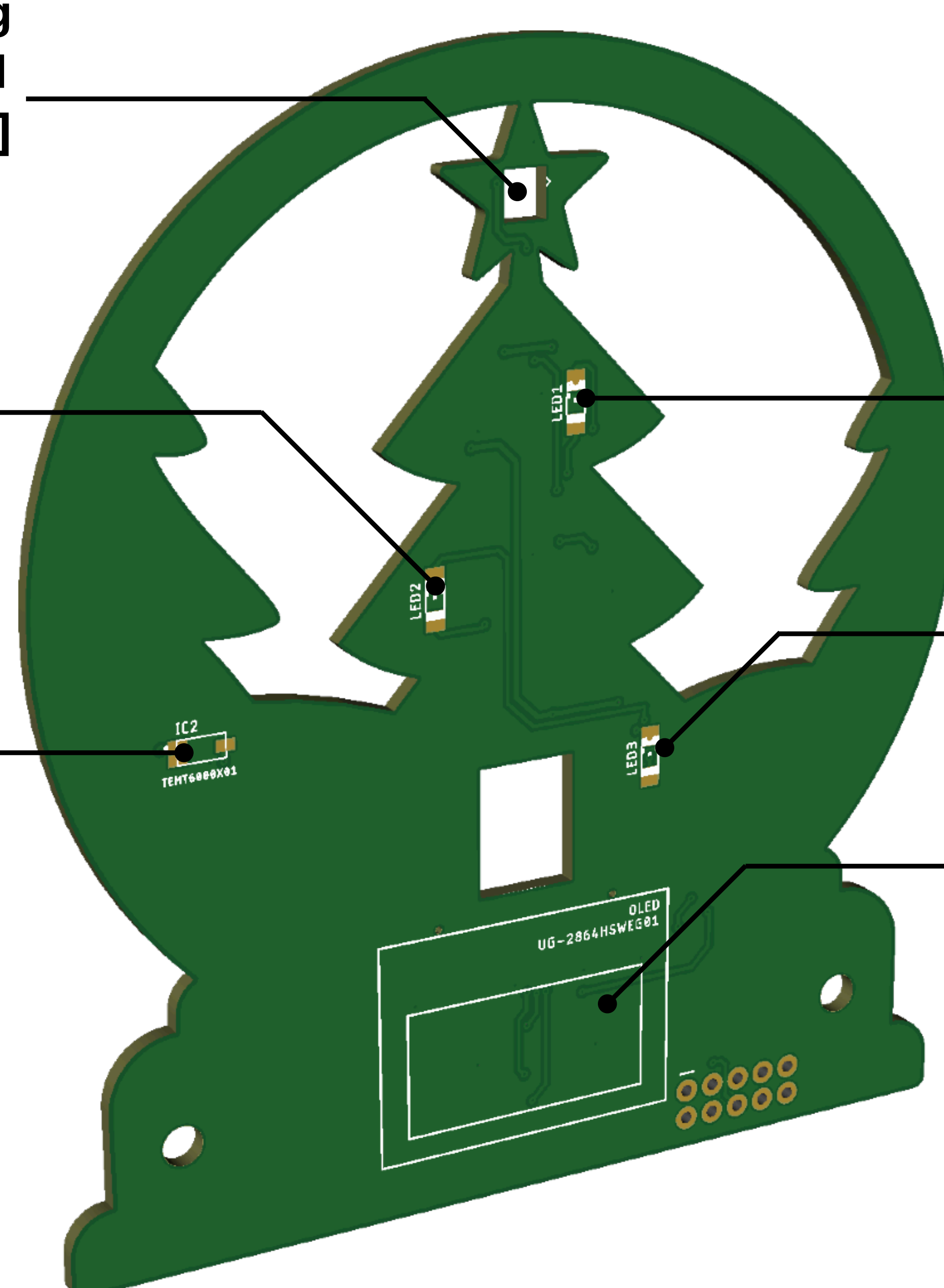
- This single board transforms your carrier board of choice, be it a micro:bit, raspberry pi Pico or the standalone WCH MCU into a fully interactive, light-up SnowGlobe centerpiece.

X’mas SnowGlobe Peripheral-PCB Major Components [Front]

[Big Star] Rear Emitting NeoPixel #1
[Pin #16]

[Baubles/Snow] LEDs #2
[Pin #14]

Analog ambient Light Sensor
[Pin #0]



[Baubles/Snow] LEDs #1
[Pin #13]

[Baubles/Snow] LEDs #3
[Pin #15]

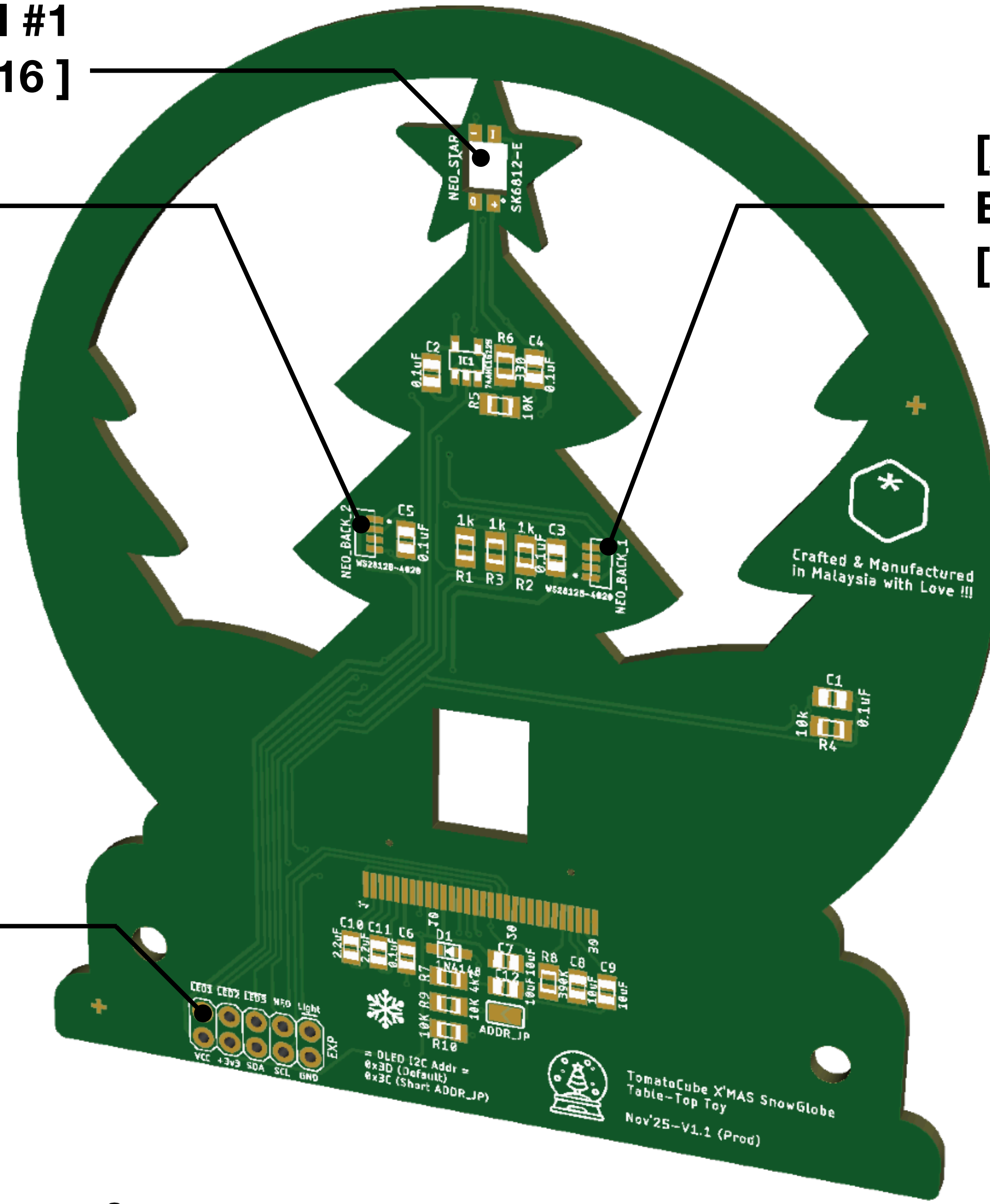
0.96” OLED Display Module [128*64]
[I2C SDA #20, SCL #19]

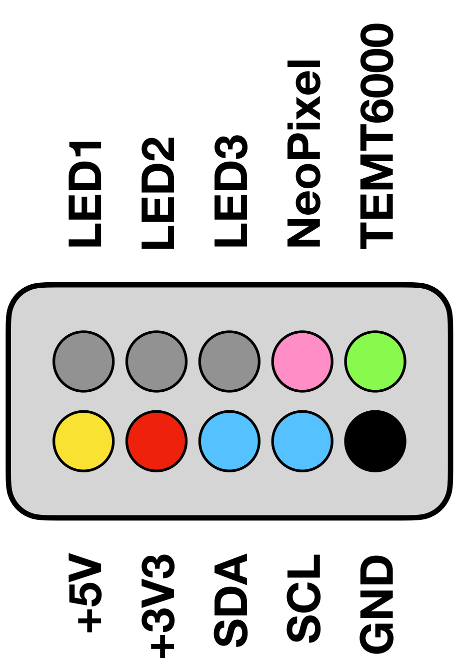
X’mas SnowGlobe Peripheral-PCB Major Components [Rear]

[Big Star] Rear Emitting NeoPixel #1 [Pin #16]

[Ambient] Side Emitting NeoPixel #3 [Pin #16]

[Ambient] Side Emitting NeoPixel #2 [Pin #16]



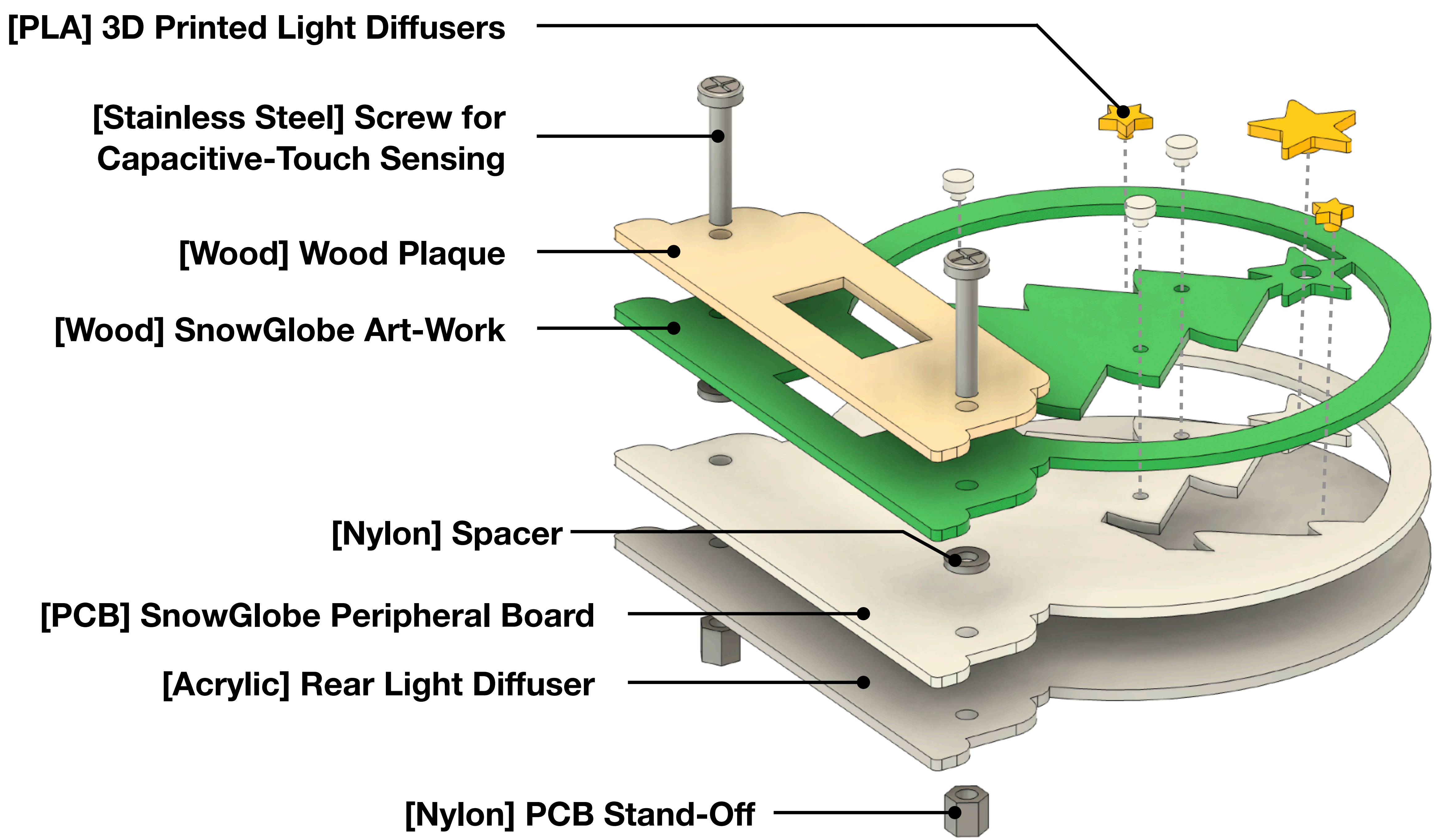


MCU/DevBoard Carrier Board (Pin Definition)

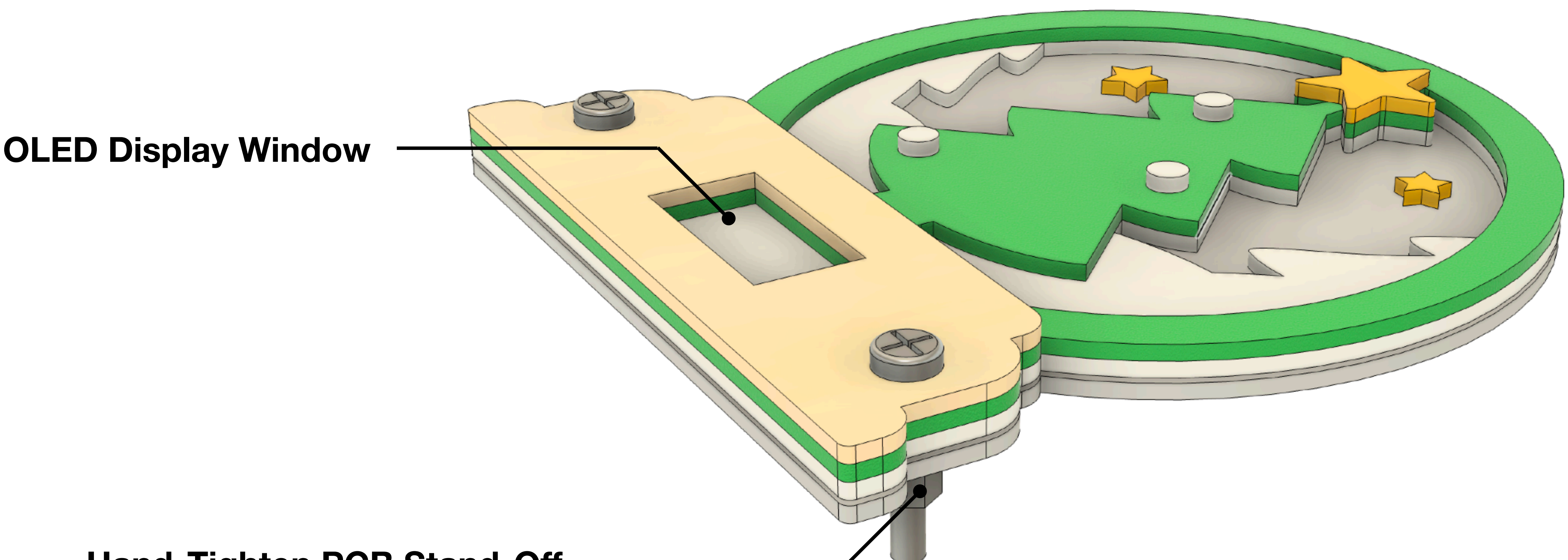
X'mas SnowGlobe Peripheral-Board Assembly

- The blown-up image/Exploded Diagram for Assembly in the next page shows how the SnowGlobe peripheral-board is arranged from top to bottom.
- **To assemble:**
    - Start from the bottom-most piece in the diagram
    - Add each layer one at a time moving upward
    - Finish by tightening the two screws and stand-offs to lock everything together
    - The diagram visually guides you, simply follow the order of layers as shown
    - Lastly, to hold everything together, we will tighten 2 screws and their respective stand-offs

X'mas SnowGlobe Peripheral-Board Components by Layer

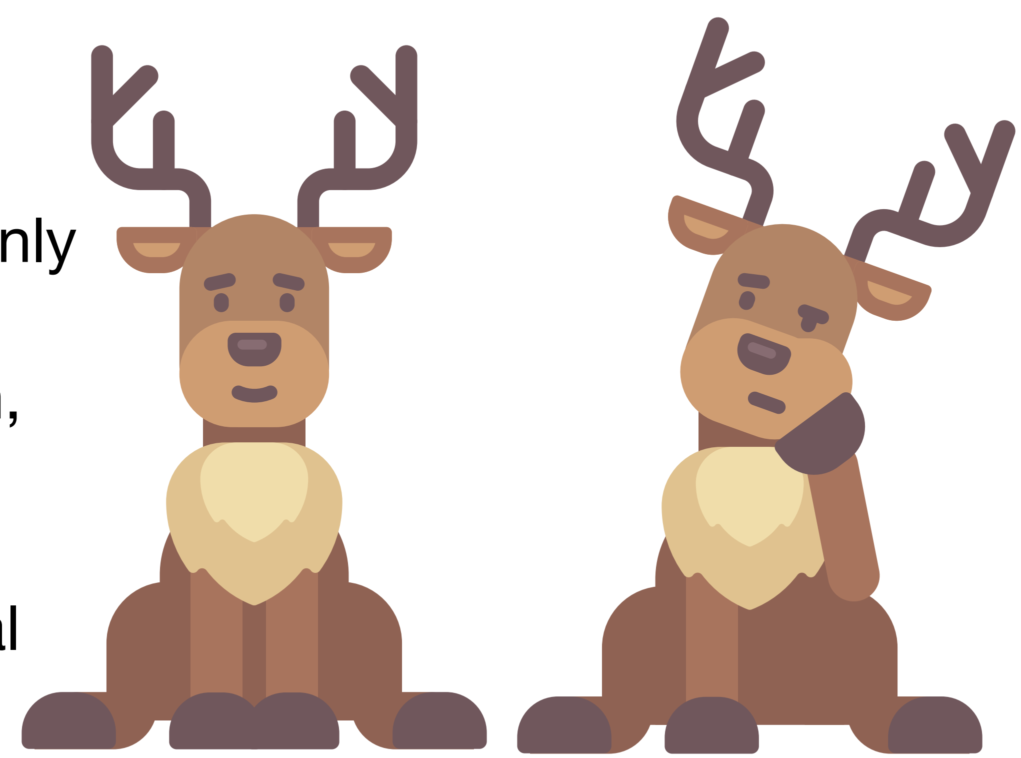


X'mas SnowGlobe Peripheral-Board Assembly Complete



Design Touches Added to Enhance the Experience

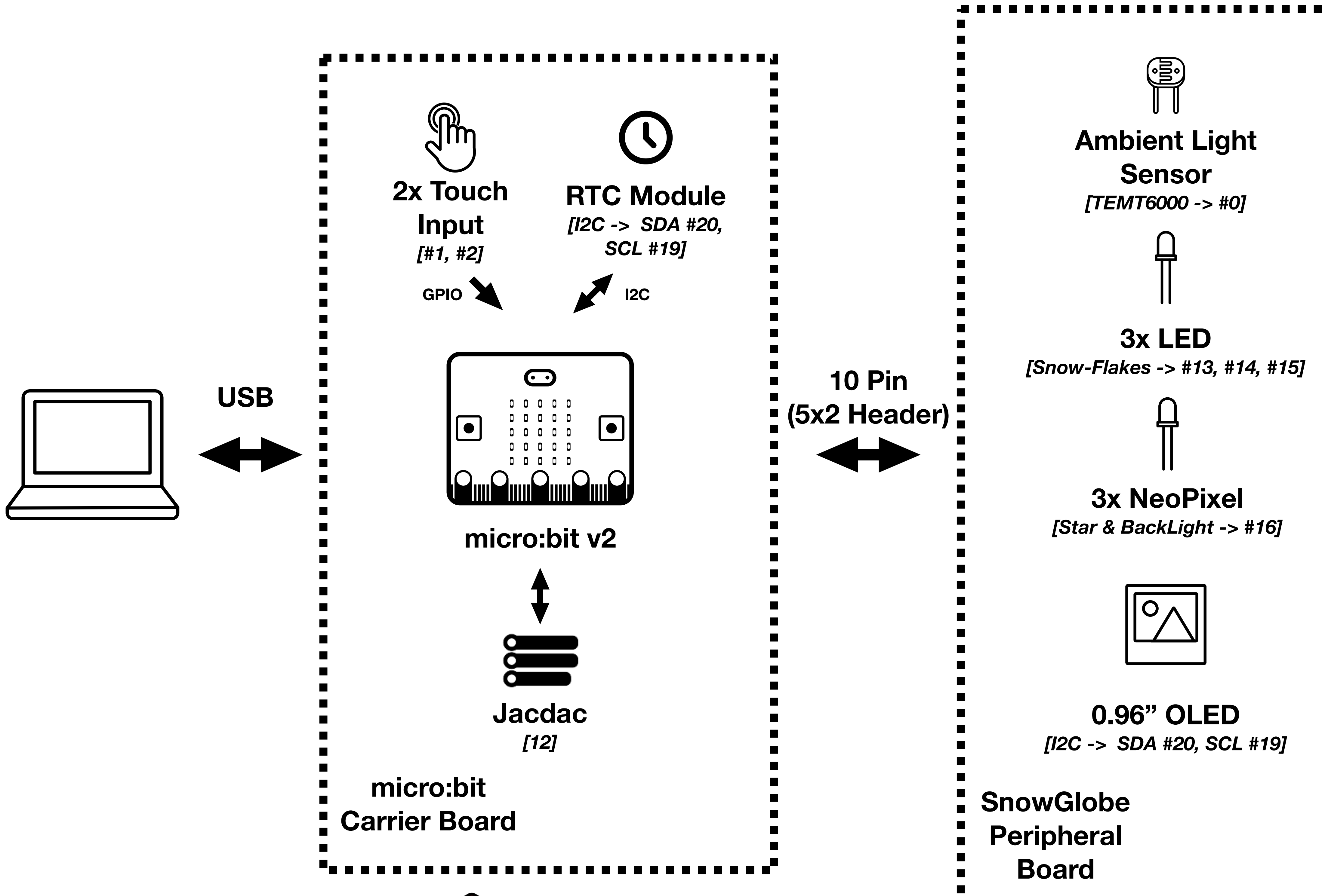
- We have painstakingly laser-cut the Wooden artwork and plaque pieces hoping to give a warm, nostalgic “real snow globe” feel
- The 3D-printed light diffusers soften LED brightness & give it a more festive glow
- Stainless-steel touch screw makes the interaction intuitive through cap-touch sensing, and it also retains the look of the SnowGlobe without intrusive Button switches
- Acrylic rear diffuser helps spread the lighting evenly
- Nylon spacers + stand-offs keep the layers clean, aligned, and sturdy
- Each layer combines electronics, craft, and visual design into one cohesive experience.



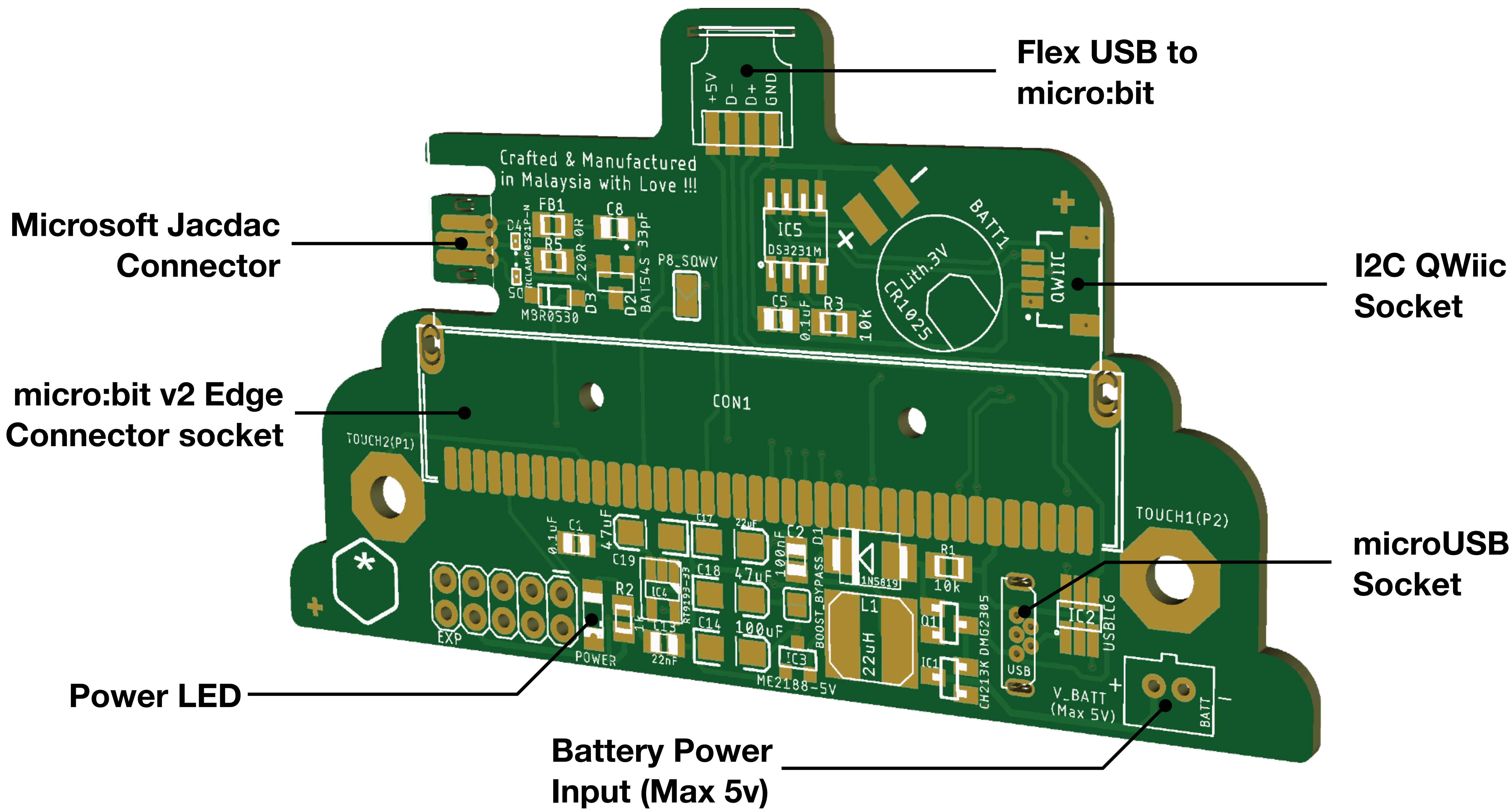
X'mas SnowGlobe Carrier Board - micro:bit v2

- The Carrier Board is the brain of the entire SnowGlobe TableTop-Toy.
  - It connects the micro:bit V2 to all the interactive features of the SnowGlobe and routes power, data, and sensors through a single, easy-to-build platform.
  - Once the micro:bit is plugged in, this board controls every animation/display, every light, every sensor reading, and every touch.
- Key Components of the Carrier Board
  - micro:bit v2 Edge Connector - The main socket that links the micro:bit
  - Battery Power Input (Max 5V) - Battery Box input supply for cordless portable operation.
  - Power LED - Indicates the board is properly powered and functioning.
  - Touch Pads (P1 & P2) - Capacitive touch inputs connected to the 2 Screws found on the Peripheral Board
  - Expansion
    - I²C Qwiic Socket - Plug-and-play connector for I²C sensors or peripherals
    - Microsoft Jacdac Connector - Supports smart accessories and future expansion via Microsoft's modular system.

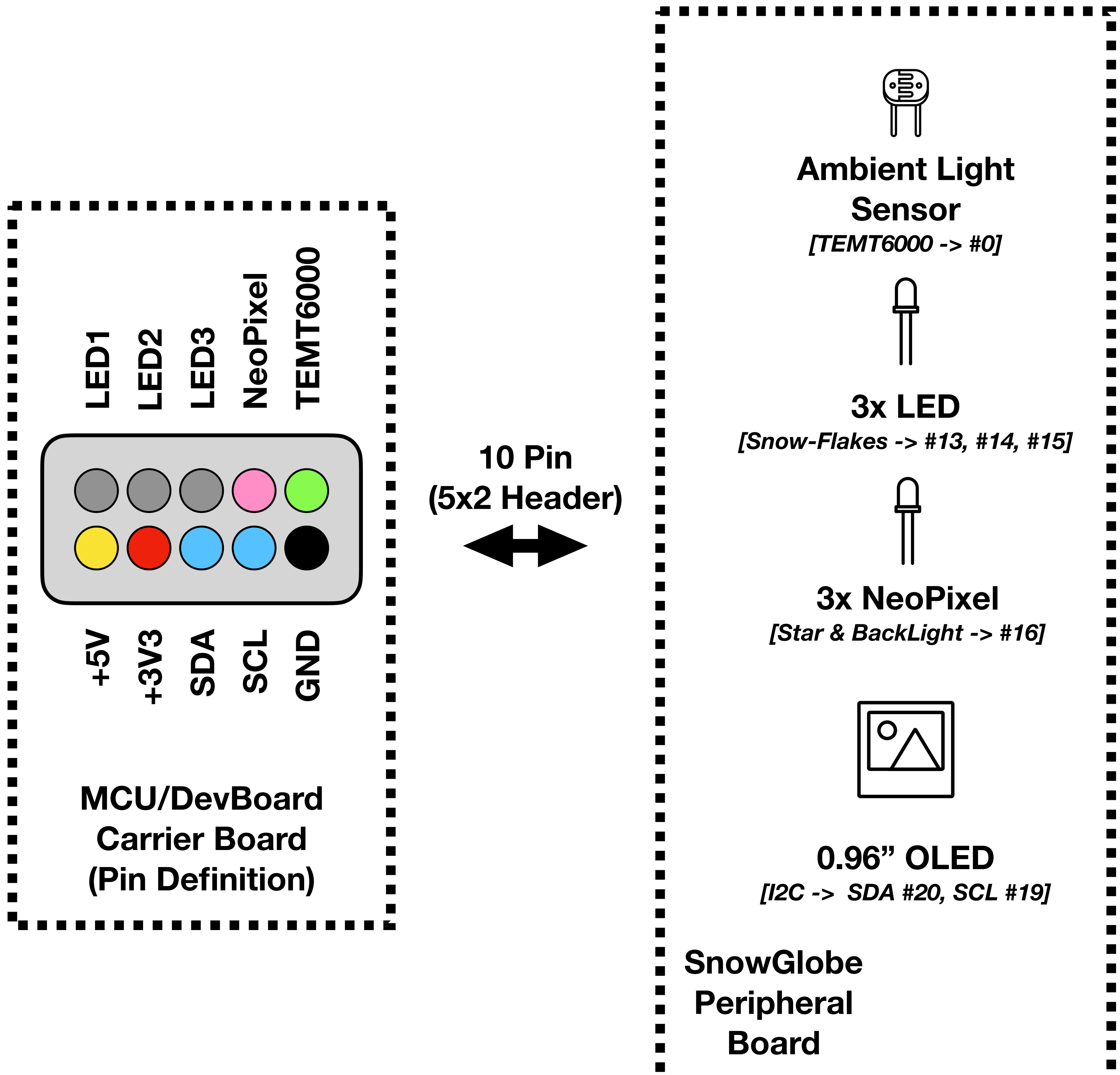
X'mas SnowGlobe TableTop Toy Carrier Board - micro:bit V2



X'mas SnowGlobe TableTop Toy Carrier Board - micro:bit V2



X’mas SnowGlobe TableTop Toy Project Kit - SnowGlobe Carrier to Peripheral Board Pinout



X’mas SnowGlobe TableTop Toy Carrier Board Assembly

- The blown-up image/Exploded Diagram for Assembly in the next page shows how the Carrier board attaches to the SnowGlobe peripheral-board.

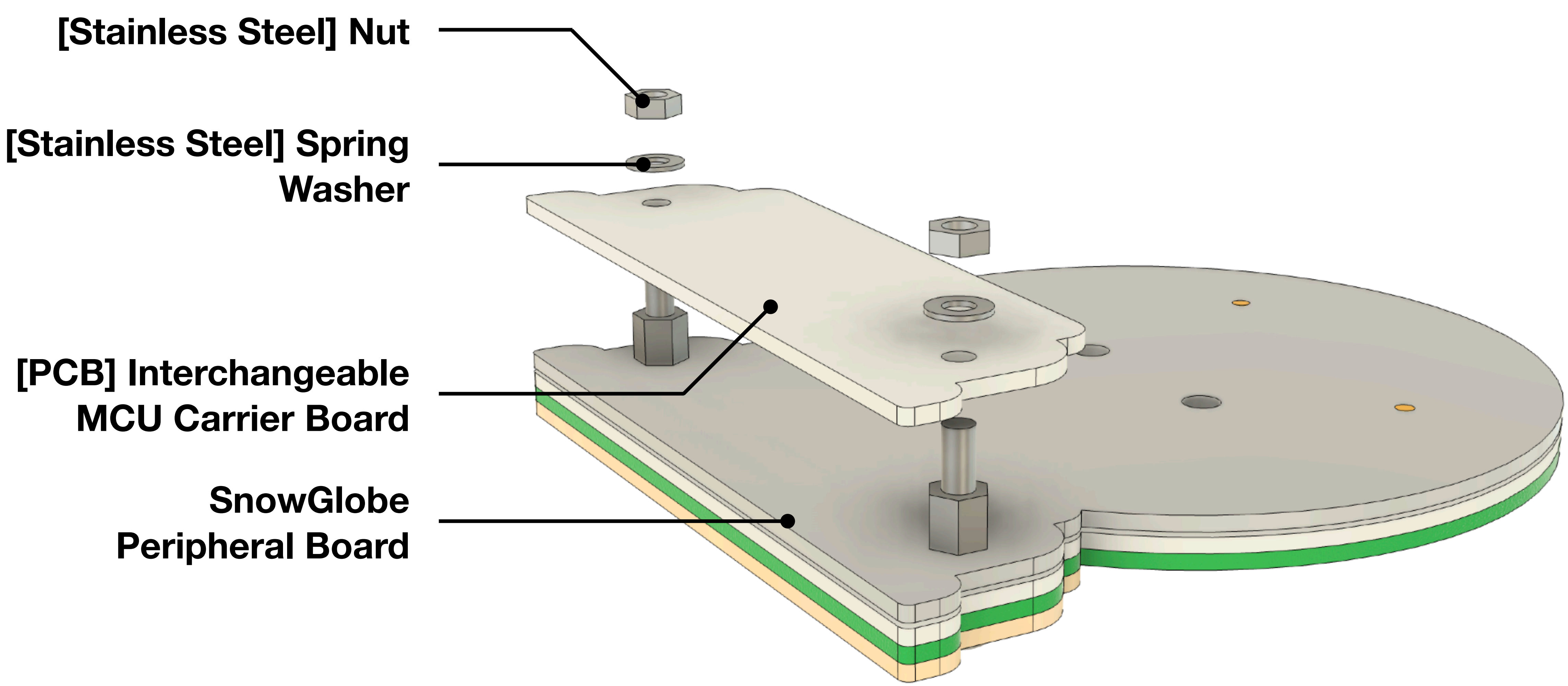
- To assemble:

  - We will now just attach the Carrier Board to the previously assembled Peripheral Board.

  - Properly align the 5x2 pin header between the Carrier board and the Peripheral board. Firmly pushes them together.

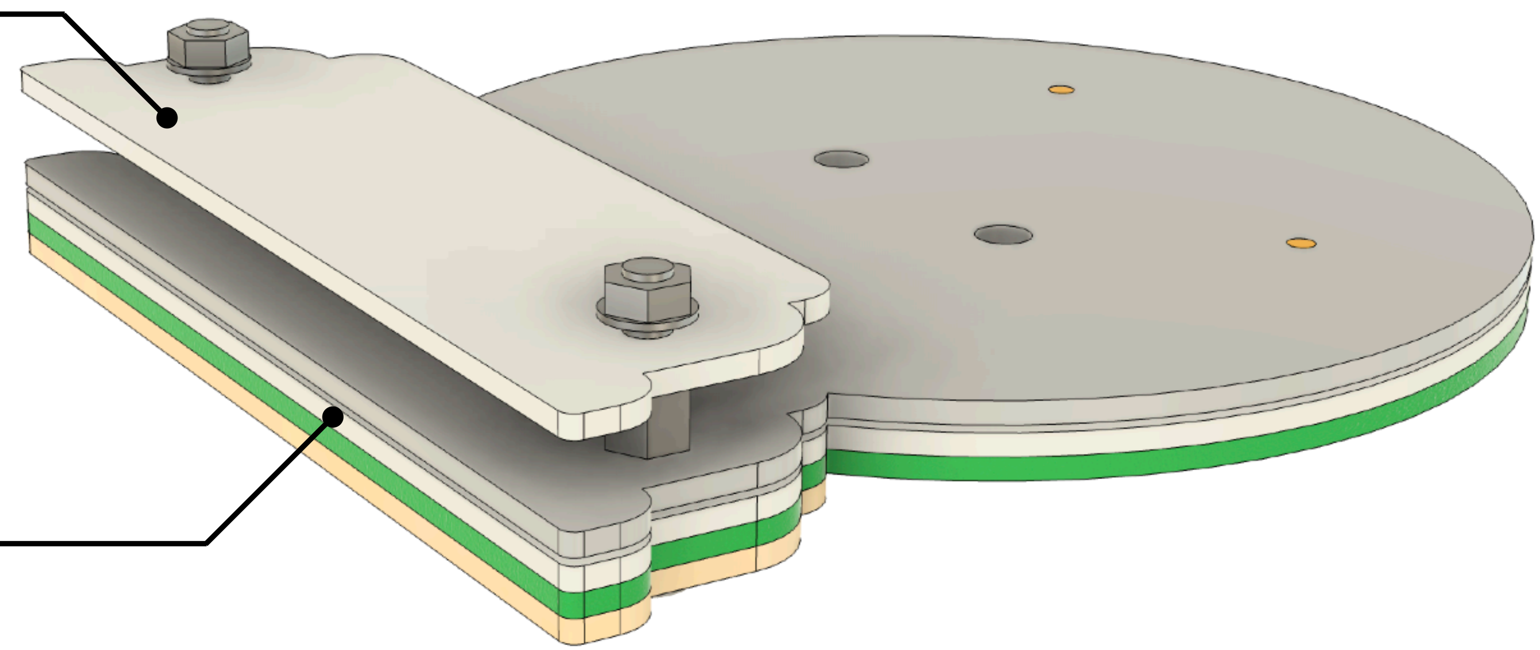
  - Using the nut & Spring Washer provided, tighten everything together, making sure that the Spring washer provide ample contact with the Capacitive Sensing pad on the Carrier PCB.

X'mas SnowGlobe TableTop Toy Project Kit - MCU Carrier-Board [V1.1]



X'mas SnowGlobe TableTop Toy Project Kit - Fully Assembled [V1.1]

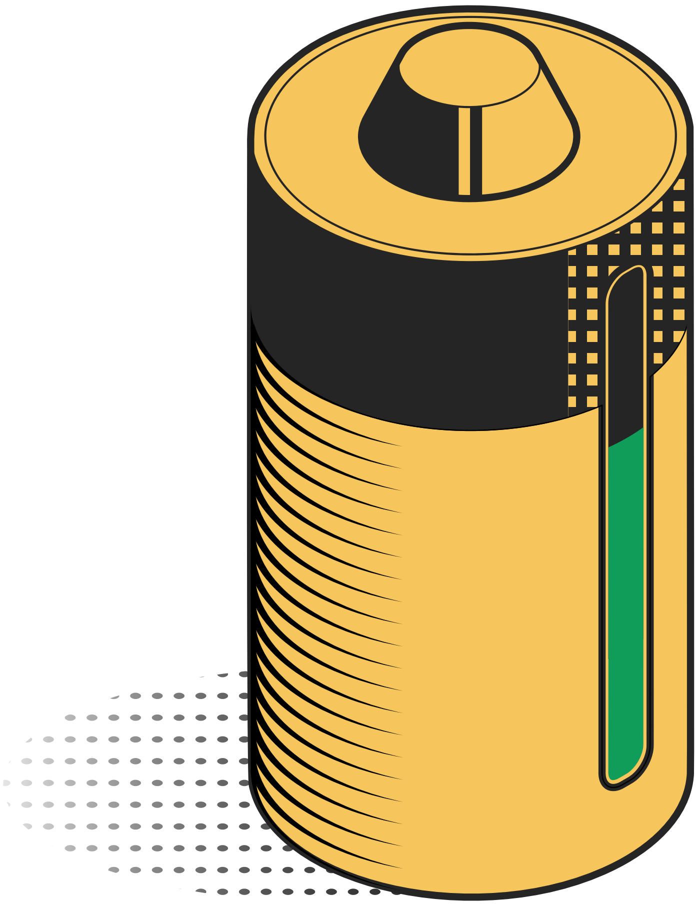
Interchangeable MCU Carrier Board
→ micro:bit v2 DevBoard
→ Raspberry Pico/W DevBoard
→ WCH CH32x033 MCU Board



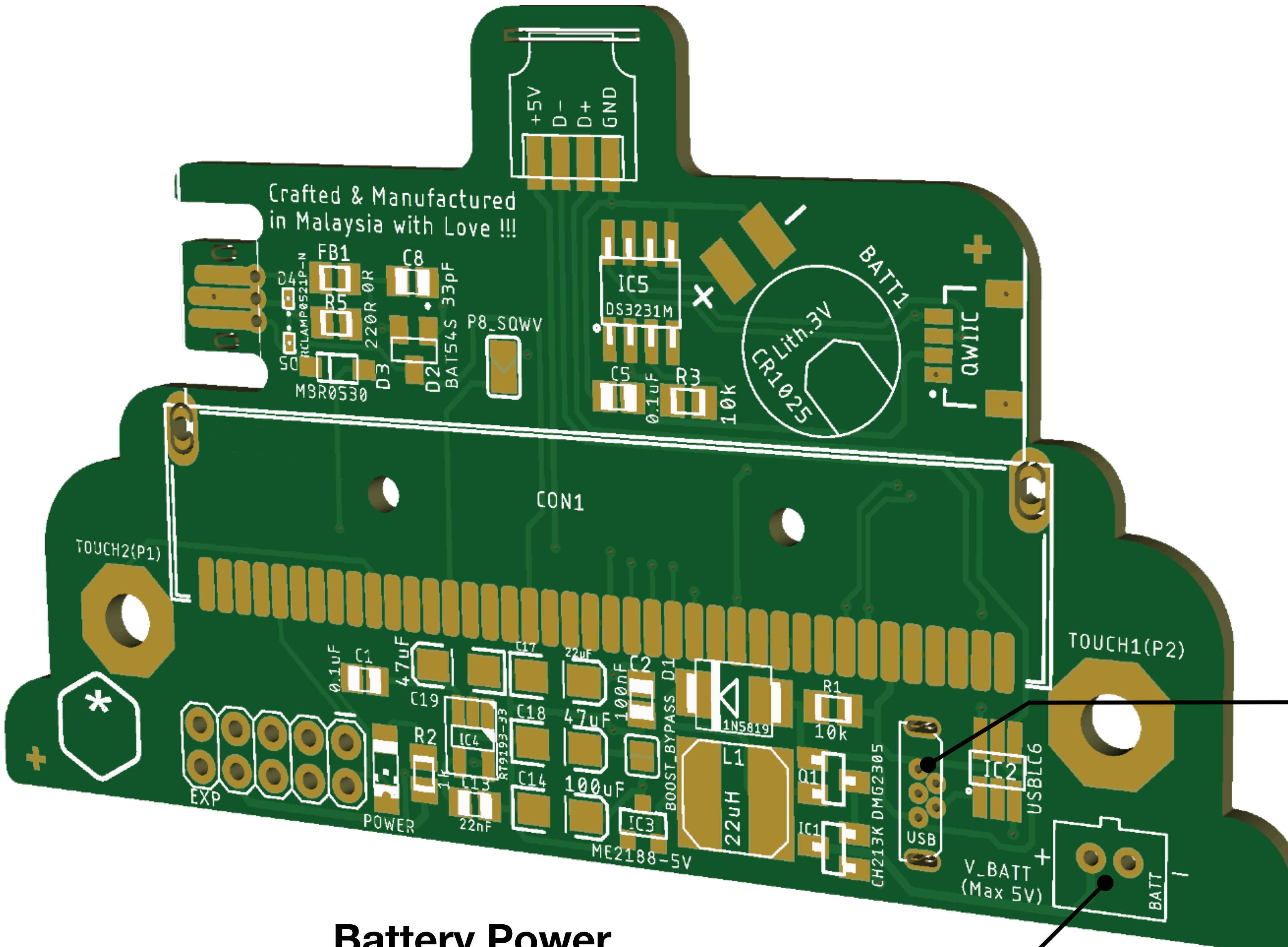
SnowGlobe Peripheral Board
- Common across multiple platform
- Reduce Cost through the use of Common Shared components

X'mas SnowGlobe Power Option

- The SnowGlobe Carrier Board supports multiple power sources, giving flexibility for both desktop and portable use:
- USB Power (Computer or Power Bank)
    - Power the board via USB using the microUSB socket.
    - Ideal for programming, testing, classroom use, and continuous animations & X'mas melodies.
    - Provides stable 5V power to both the carrier board and the micro:bit via the Flex USB Connector.
- 2×/3x AA Battery Pack
    - Connect the included AA battery box for portable, self-powered operation.
    - Great for displays, events, or locations where USB power isn't available.
    - Keeps the SnowGlobe running without cables or external devices.



X'mas SnowGlobe TableTop Toy Carrier Board - micro:bit V2 - Power Options



microUSB Socket

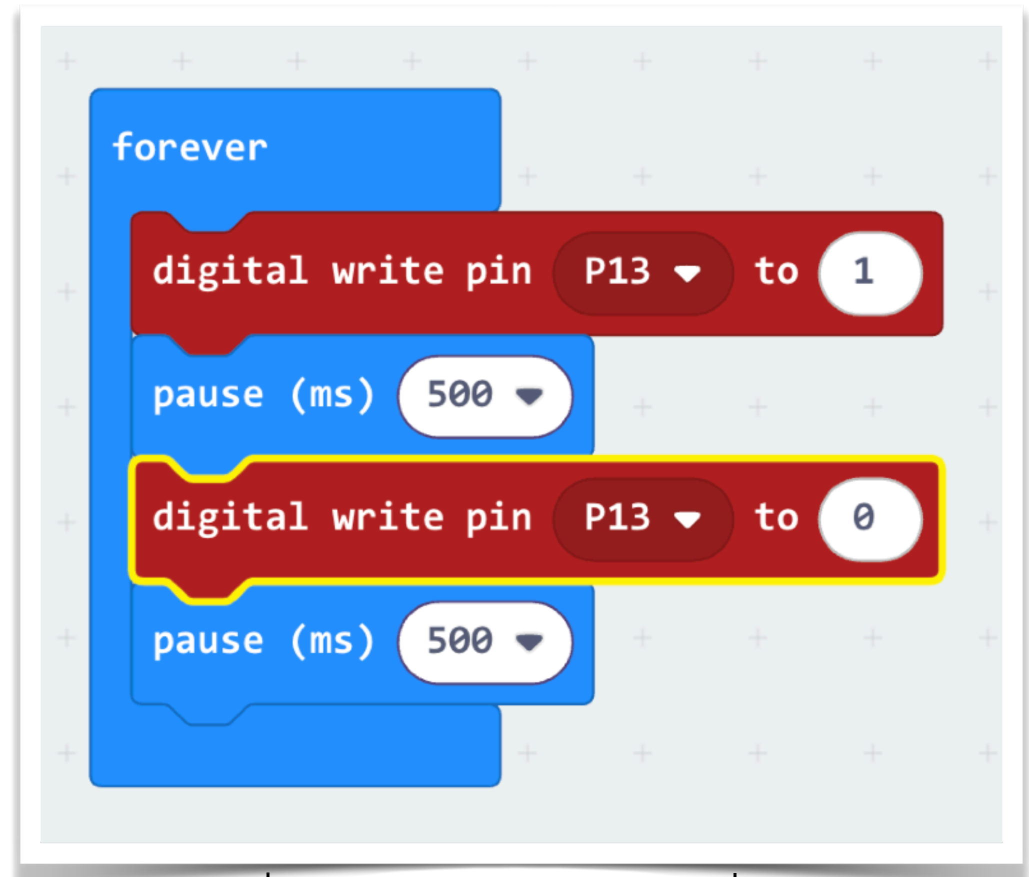
Battery Power Input (Max 5v)

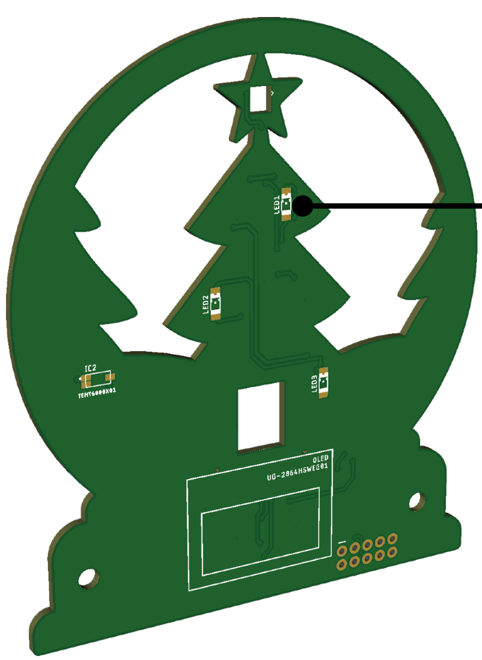
- Part 3 -

Carrier Board programming (micro:bit V2)

“Hello World” of embedded world - Blink

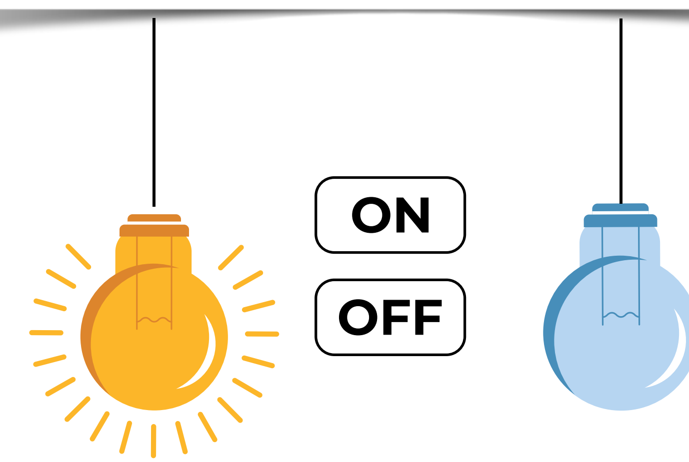
- The very first program in embedded systems is usually Blink - making an LED turn on and off. Just like **“Hello World”** in programming, Blink introduces the core ideas that every embedded project builds on:
    - Controlling hardware using code
    - Timing and delays
    - Understanding digital outputs
    - Seeing immediate physical feedback
- With the SnowGlobe and micro:bit, the Blink program will flash the Top LED Bauble/Snow - Your first step into the world of embedded electronics and interactive devices!





[Baubles/Snow] LEDs #1
[Pin #13]

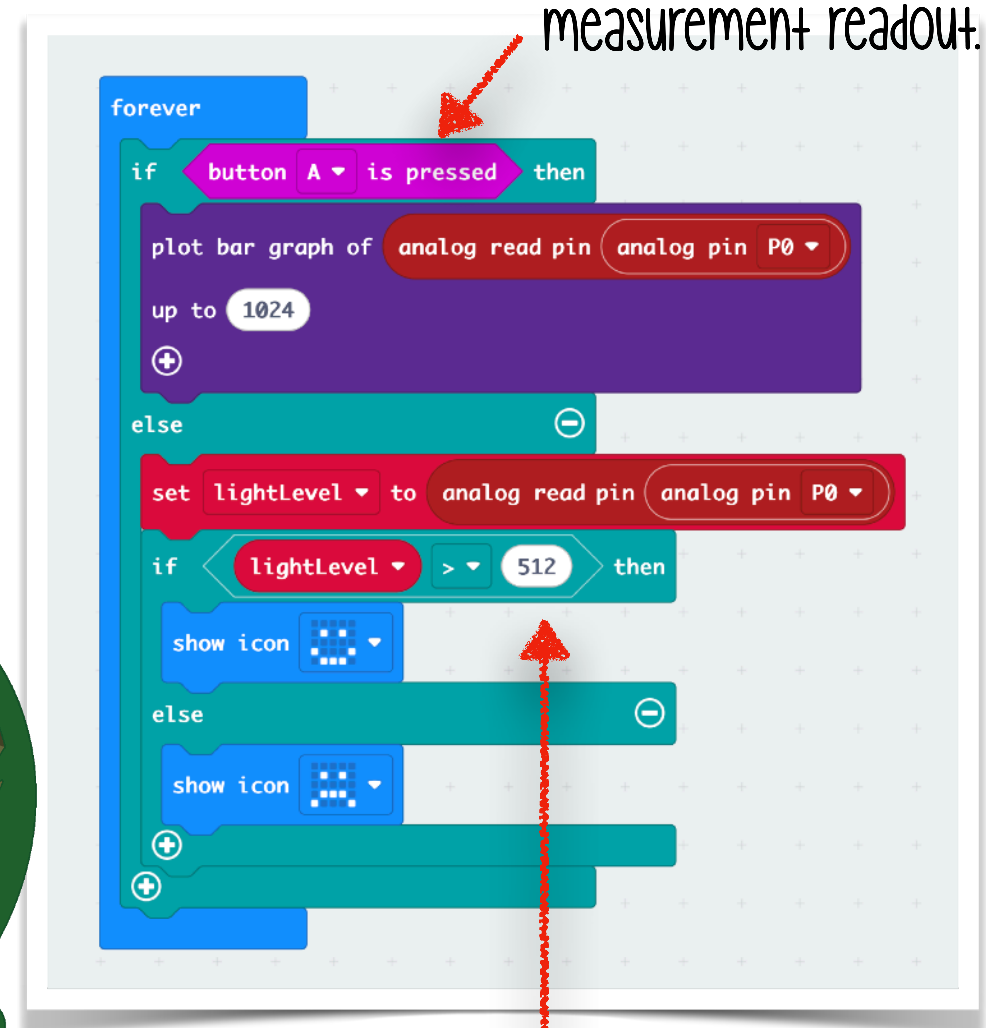
LED is connected to pin #13



“Let there be Light” - Detecting ambient light intensity

Mr. Vampire is afraid of the bright sun, he needs a measuring tool to know if it is okay to venture out into the open.

Pressing Button A will get a Bar-graph display of the light measurement readout.



Setting the brightness threshold value

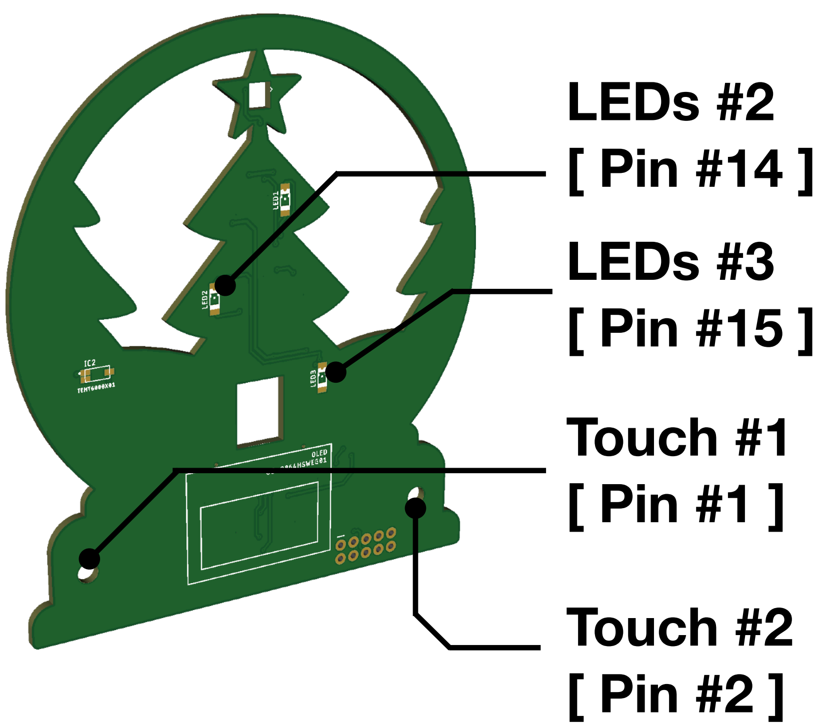
The SnowGlobe uses an Analog Light sensor on the Peripheral board to detect the light level. In **shadow**, it reads the light level as 0; whenever it senses light, it reads more than zero, the light level ranges from **0 to 1024**.

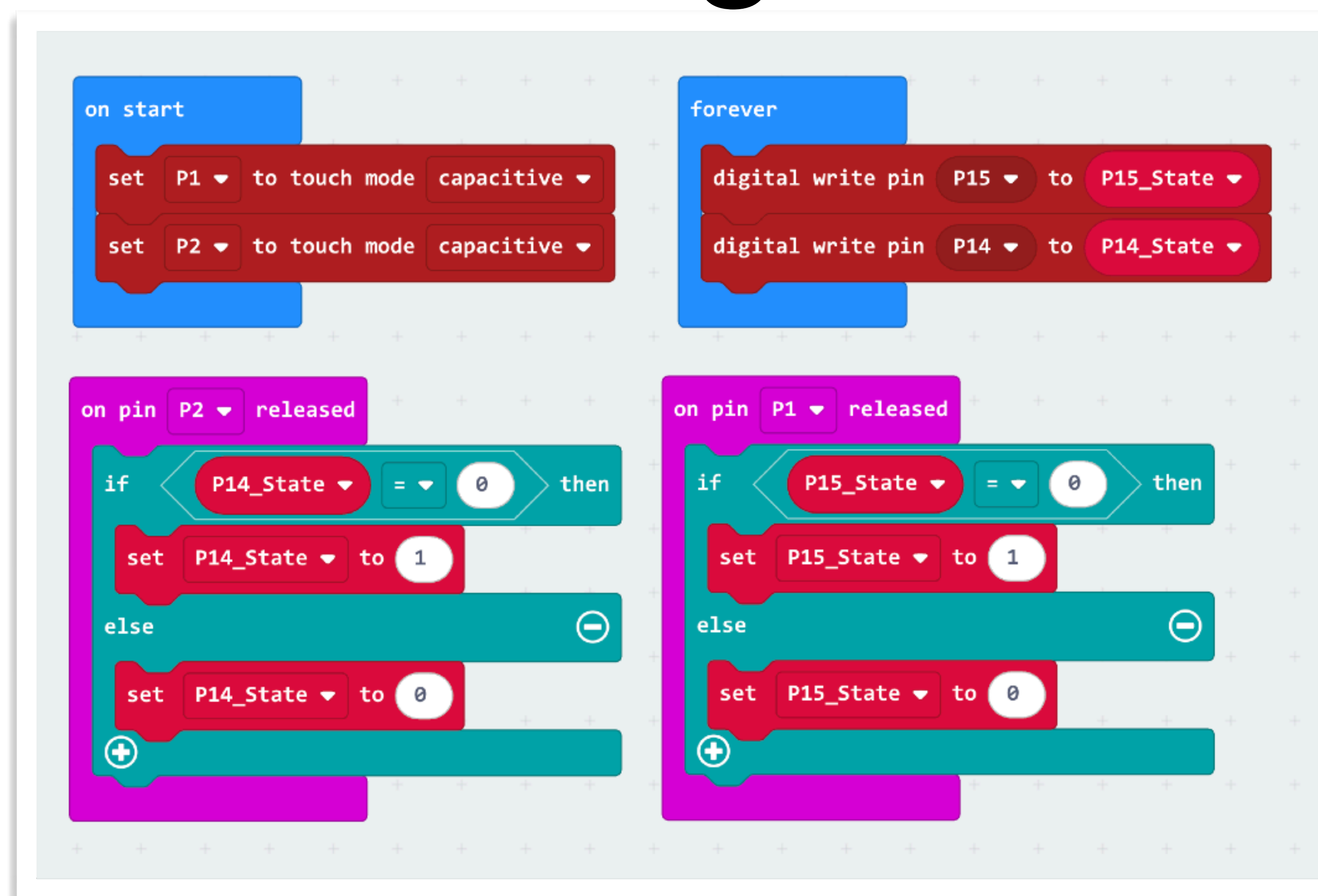
Basically it means that the light level shows how bright or dark the place where our SnowGlobe is placed.

Light sensor connected to Pin #0

Interaction - Detecting Touch

Touch sensor Pad connected to Pin 1 & 2





- The micro:bit lets us use certain pins - like P1 and P2, as capacitive touch inputs. This means we can detect when a student lightly touches the touch sensitive screws with a finger.
- Your body acts like a small capacitor, and the micro:bit senses this tiny change in electrical charge. In this activity, touching the touch sensitive screws will toggle the state of two LEDs on the board. This creates a simple, intuitive interaction:
  - Touch Sensitive screw → micro:bit senses change → LED toggles ON/OFF

“Hark the herald angels sing” - Playing a melody 🎶

Everyone **loves** making sound (noise) 🖤

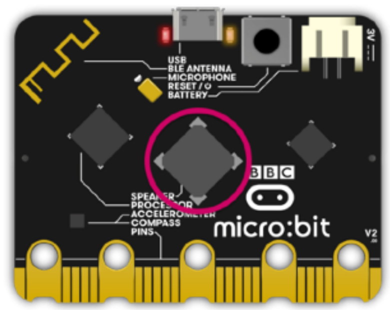
Sound & music is produced when an object **vibrates**, creating a varying pressure wave. This pressure wave causes particles in the surrounding medium, such as air around us, to have vibrational motion.

As the particles vibrate, they move nearby particles, propagating the sound further & further away through the medium. 🔊

Annoy your parents (Generating Sound)

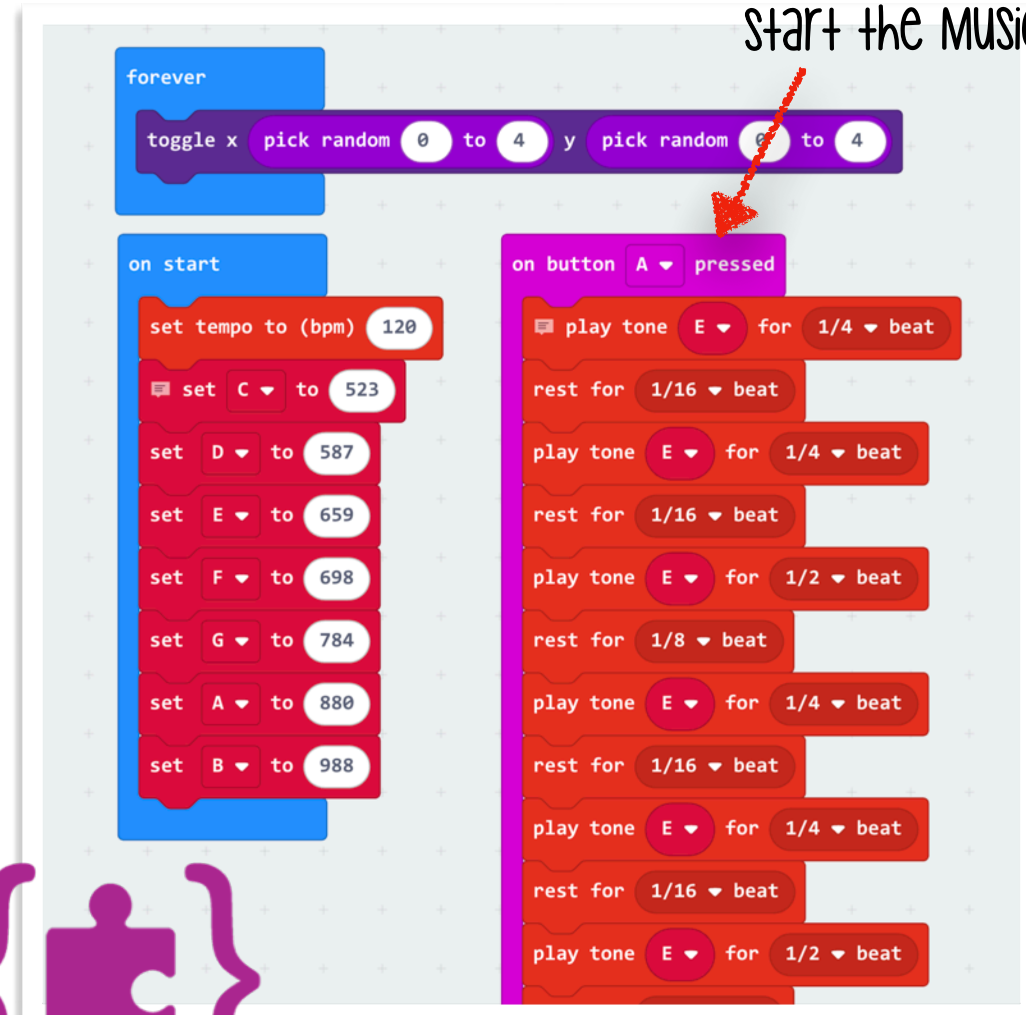
Pressing Button A will start the music

- The micro:bit v2 can play music by sending tones to a **onboard buzzer/speaker.**
  It does this by generating different frequencies pulses, each representing a musical note.



- By combining notes with timing, we can create melodies, sound effects, or even full songs.

- In this activity, we'll make the micro:bit play a short tune from a popular Christmas jingle.



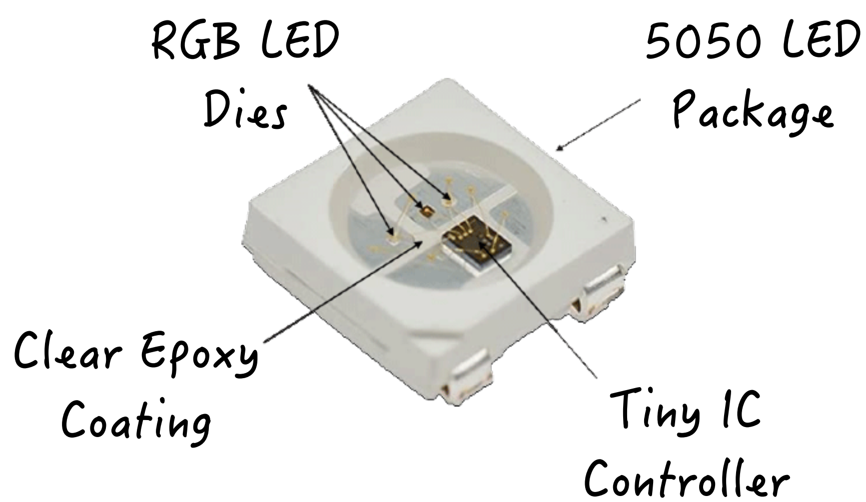
Click on the MakeCode icon to load the sample code found on the slide !!

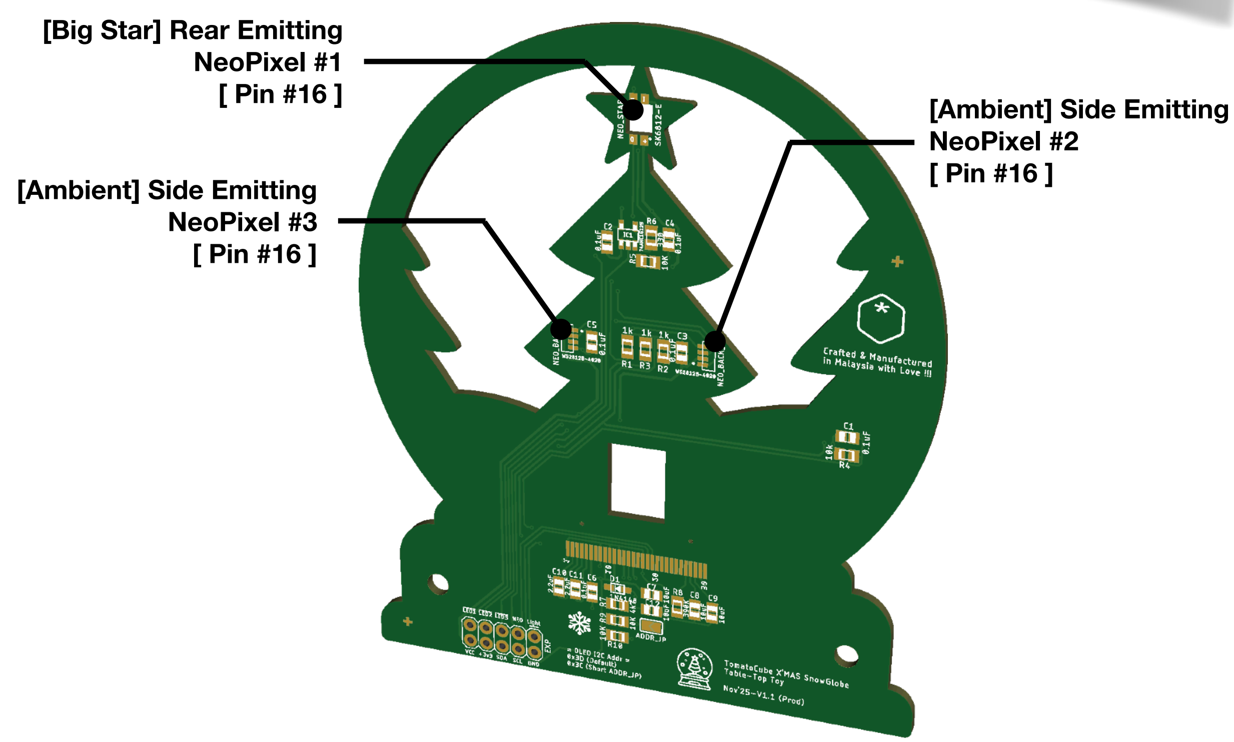
Controlling the NeoPixels (RGB LEDs)

NeoPixels are connected to Pin 16

A NeoPixel is a chain-able, individually addressable, color-changing LED that is controlled via a single pin on the micro:bit.

Within our SnowGlobe project kit, you will find 3 NeoPixels, one as the "Big Star" and two as the Ambient lighting diffused by the rear Acrylic piece.



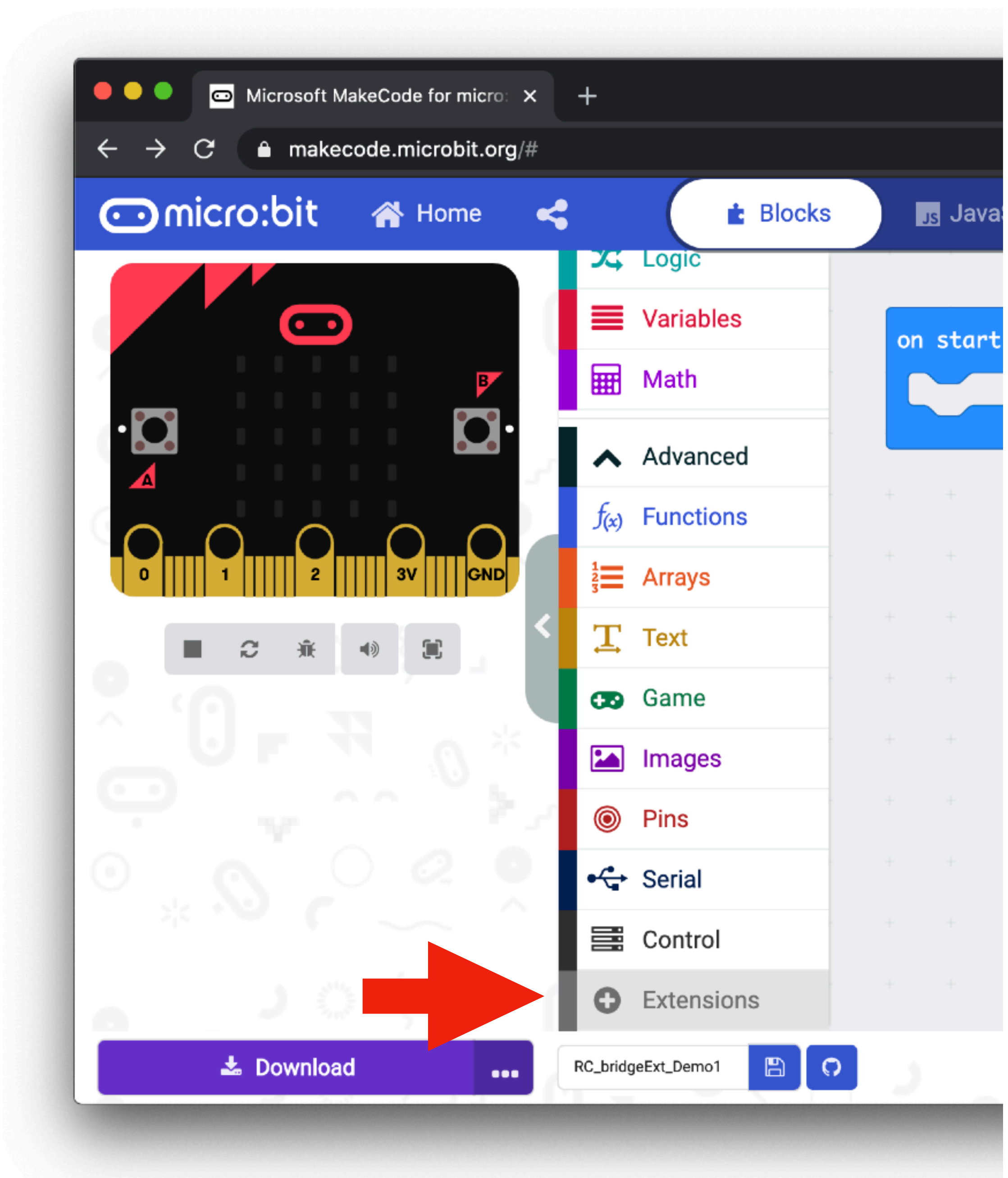


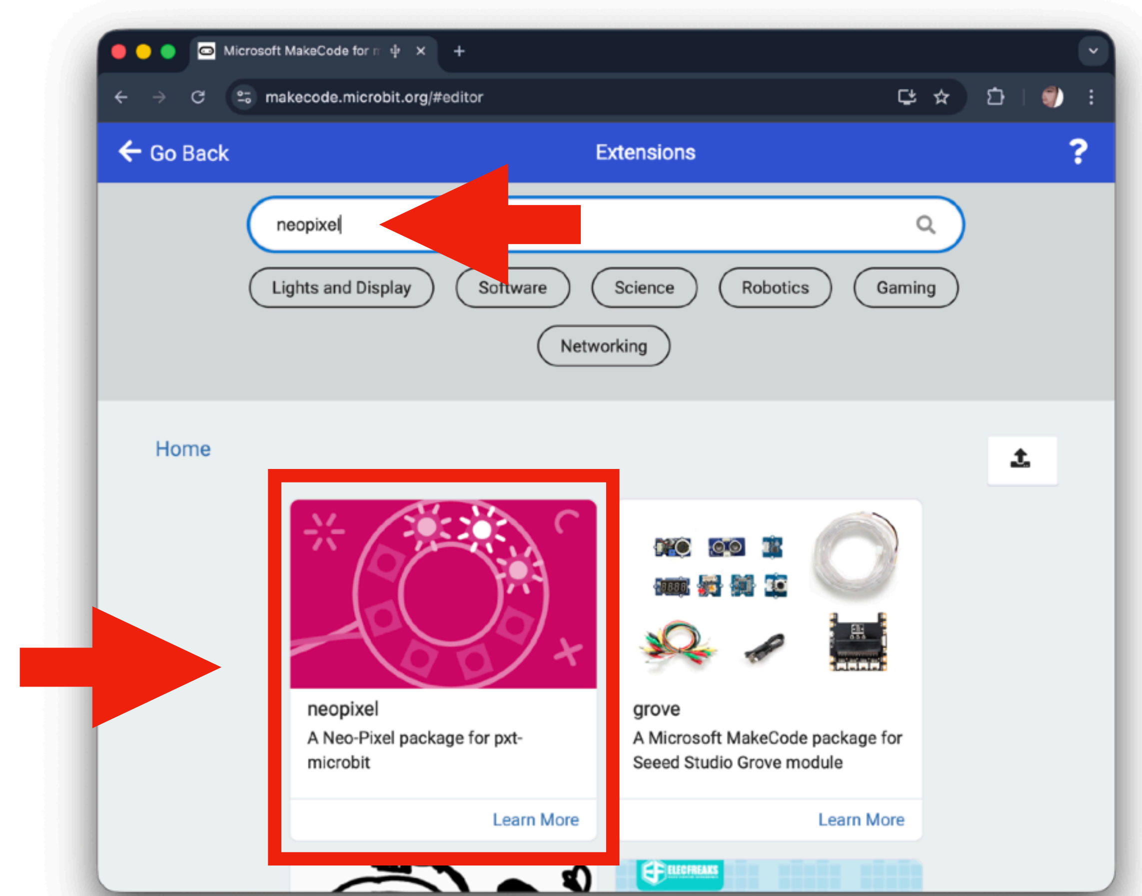
Installing additional MakeCode® extensions

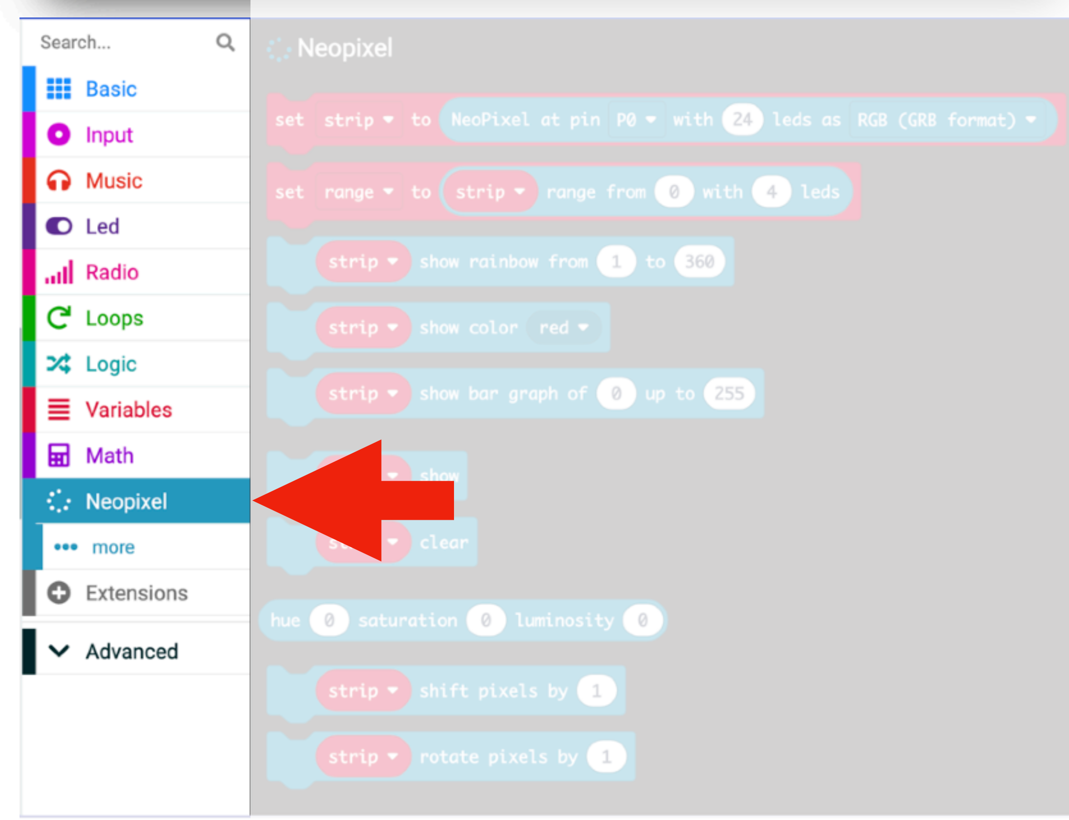
Right out of the box, Microsoft® Makecode supported lots of off-the-shelf hardware.
But it cannot cover everything under the **Sun**.

To facilitate interoperability with custom hardware add-on. We need to add some additional MakeCode PXT extension to our micro:bit MakeCode development environment.

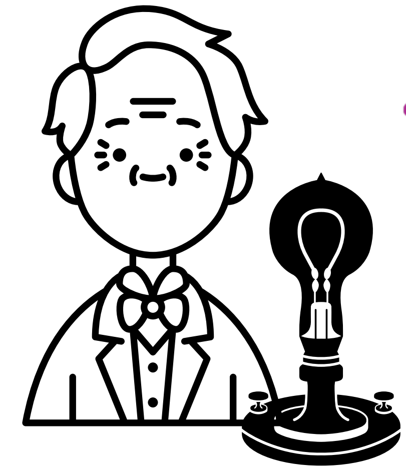
Adding NeoPixel MakeCode extension





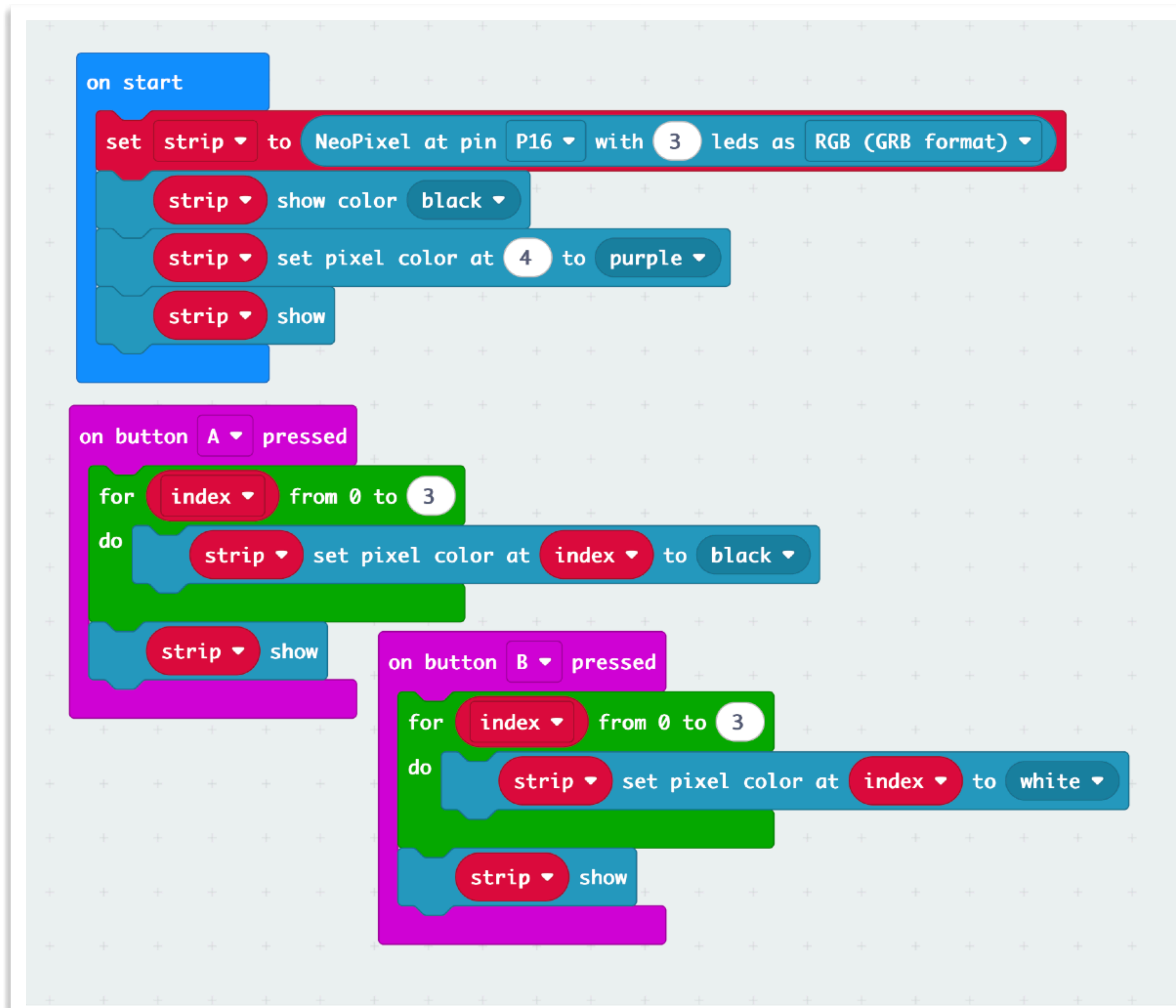


Controlling the Neopixels (RGB LEDs)



Neopixels are connected to Pin 16

- In this demonstration, we are controlling the NeoPixel LEDs using our micro:bit & the Capacitive Touch capability demonstrated earlier.
- We will turn on the **First** NeoPixel LED (Big Star) to a color of **Orange**.
- Pressing button A or B will **toggle** the back-light of the SnowGlobe, the **2nd & 3rd NeoPixel LEDs**.



Knowing & Showing Time - OLED & RTC

- What is an **OLED** display?
  - OLED is Organic Light Emitting Diode where each pixel emits light in response to an electric current.
  - Unlike LCD screens that need a backlight, OLED pixels glow individually giving:
    - Sharp, high-contrast text
    - Very dark blacks (pixels actually turn off)
    - Low power consumption
    - Good viewing angles

- Why do we use it in micro:bit projects?
  - OLED modules are small but powerful. They let us display:
    - Sensor readings (temperature, light level, etc.)
    - Icon animations
    - Text messages
  - Because the micro:bit has no built-in screen (other than the 5×5 LED grid), the OLED becomes a portable “information window” for our projects.

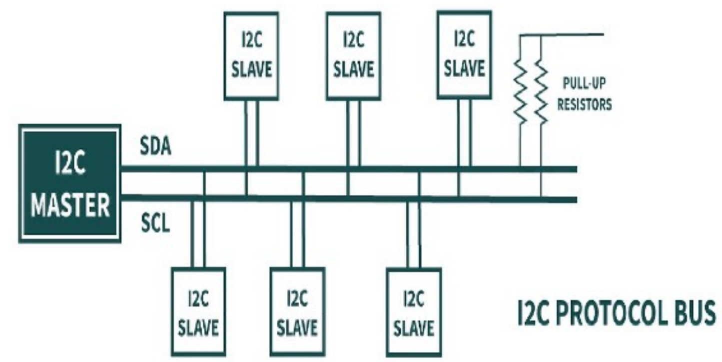
- Specification of OLED Screen on our SnowGlobe Board:
  - Display Size: **128x64 pixels**
  - Display Driver: **ssd1306**
  - Display Colors: Monochrome
  - Interface: I2C
  - Operating Current: ~20mA

OLED Display
[I2C SDA #20, SCL #19]

About the Real Time Clock (RTC) Component

- What is an **RTC**?
  - A Real Time Clock (RTC) is a small chip that keeps the current Time & Date
  - The RTC chip found on the SnowGlobe is the **DS3231**, an accurate, stable, and has a built-in temperature-compensated crystal.
- Why do we need an RTC?
  - The micro:bit does not keep the correct time when powered off. Every time you reboot, it forgets what time it is.
  - The RTC solves this problem by:
    - Having its own tiny battery
    - Running continuously even when the micro:bit is off
    - Maintaining very accurate time (drift only a few minutes per year)

- How the OLED Display and the RTC connects to the micro:bit
  - Both the OLED and the RTC uses I^2C communications, a communication protocol shared on two pins:
    - SDA → Data line
    - SCL → Clock line
  - Because I^2C supports multiple devices on the same two wires, both the OLED and RTC can work together on a single micro:bit.
  - This makes peripherals utilizing the I^2C protocols, easy & pin economical to use with just 2 wires.



Installing 2 more MakeCode® extensions

- Before we can build our working clock for the Snow Globe tabletop project, we need to install two custom MakeCode extensions. These extensions will add new blocks that let the micro:bit to Display graphics on the **OLED screen** & Communicate with the **RTC IC** for accurate timekeeping

- Summary for the next steps:

  - **OLED12864_I2C Extension**

    - This extension gives us all the blocks we need to draw text and graphics on the OLED display.

    - Extension URL: **https://github.com/makecode-extensions/OLED12864_I2C**

  - **TomatoCube SnowGlobe Extension**

    - This extension is designed by us to:

    - Rotate the OLED content upright

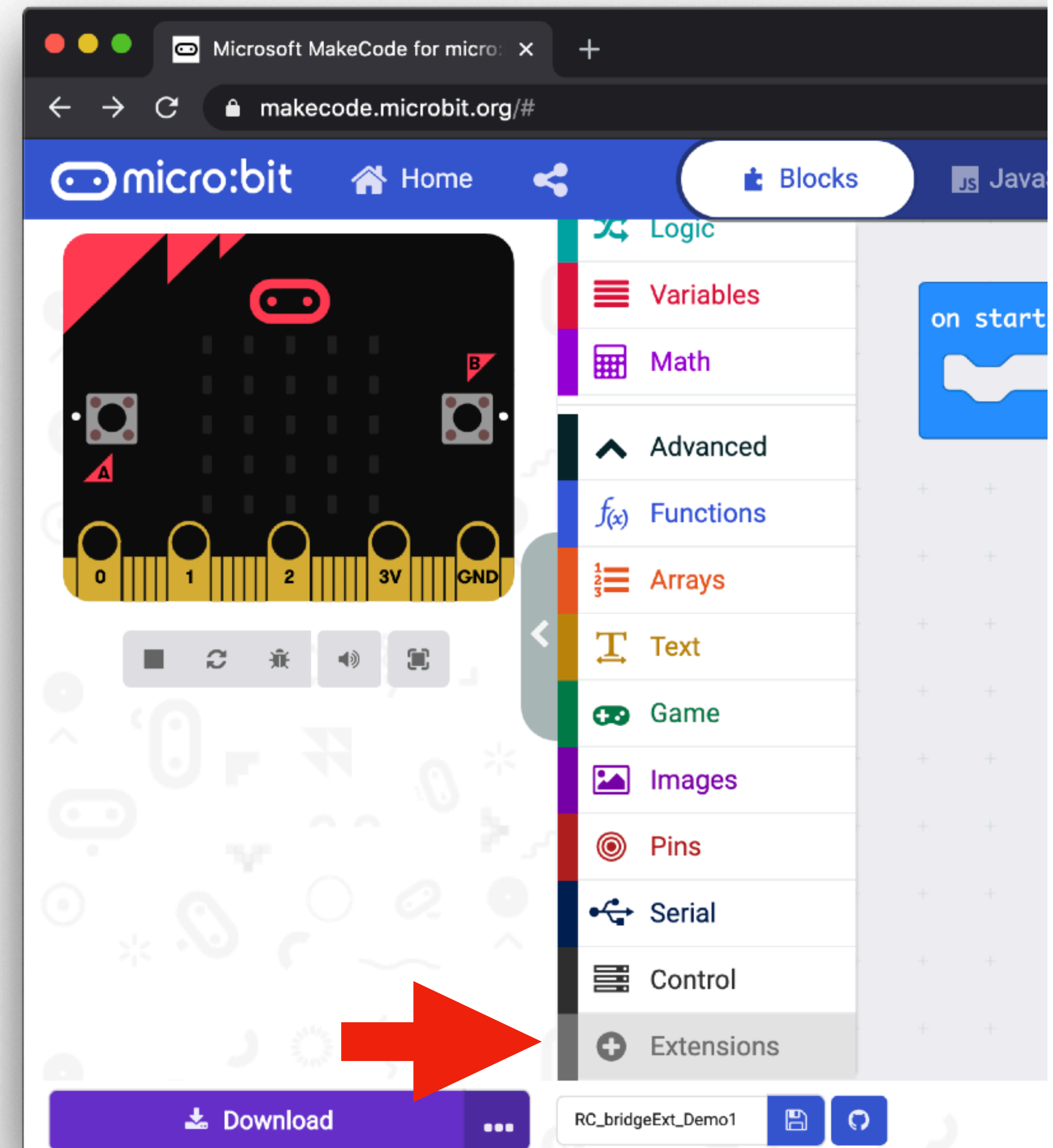
    - Provide helper blocks for the RTC clock

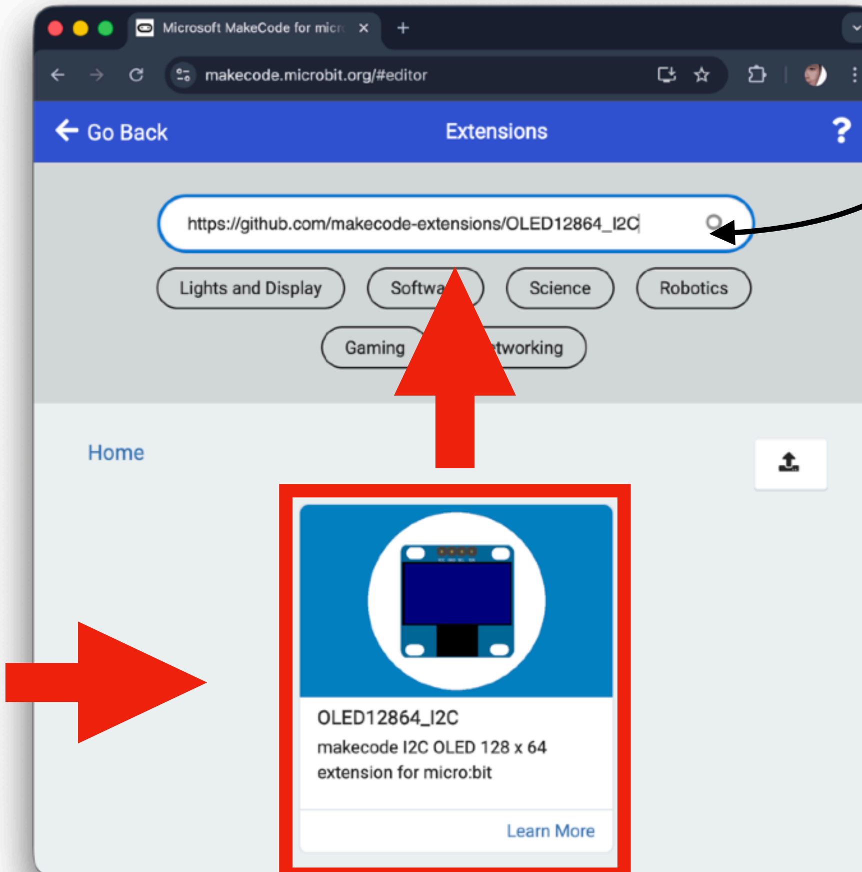
    - Extension URL: **https://github.com/TomatoCube18/pxt-tc-snowglobe**

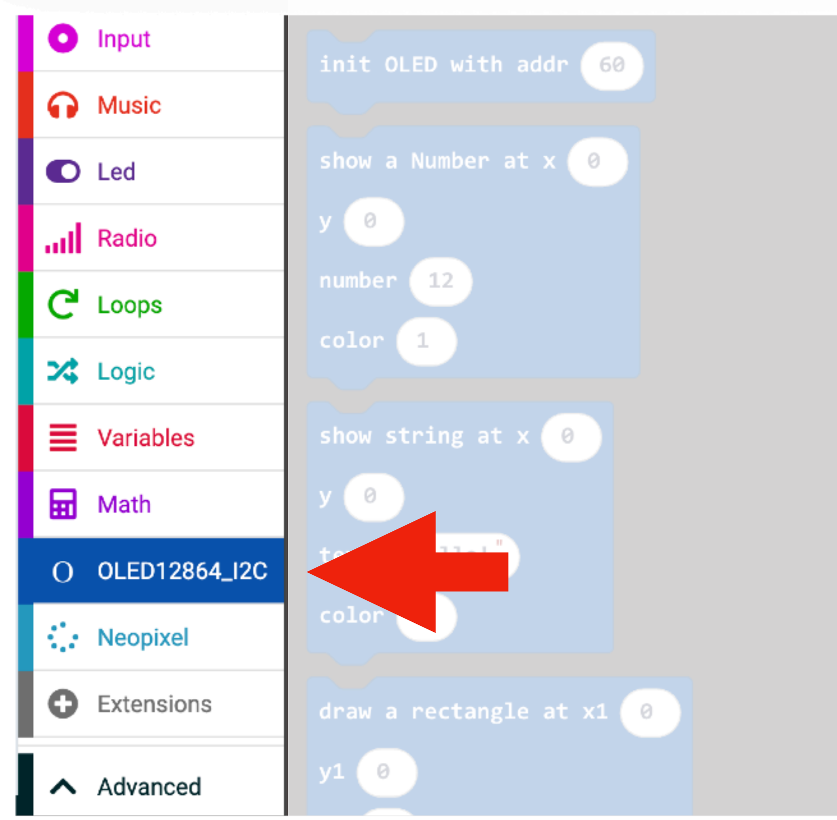
!! Important Tip for Students !!
To avoid typos, copy and paste the URLs directly from this slide (PDF).
MakeCode will only find the correct extension if the link is typed exactly right.

https://github.com/makecode-extensions/OLED12864_I2C

Adding OLED Display MakeCode extension

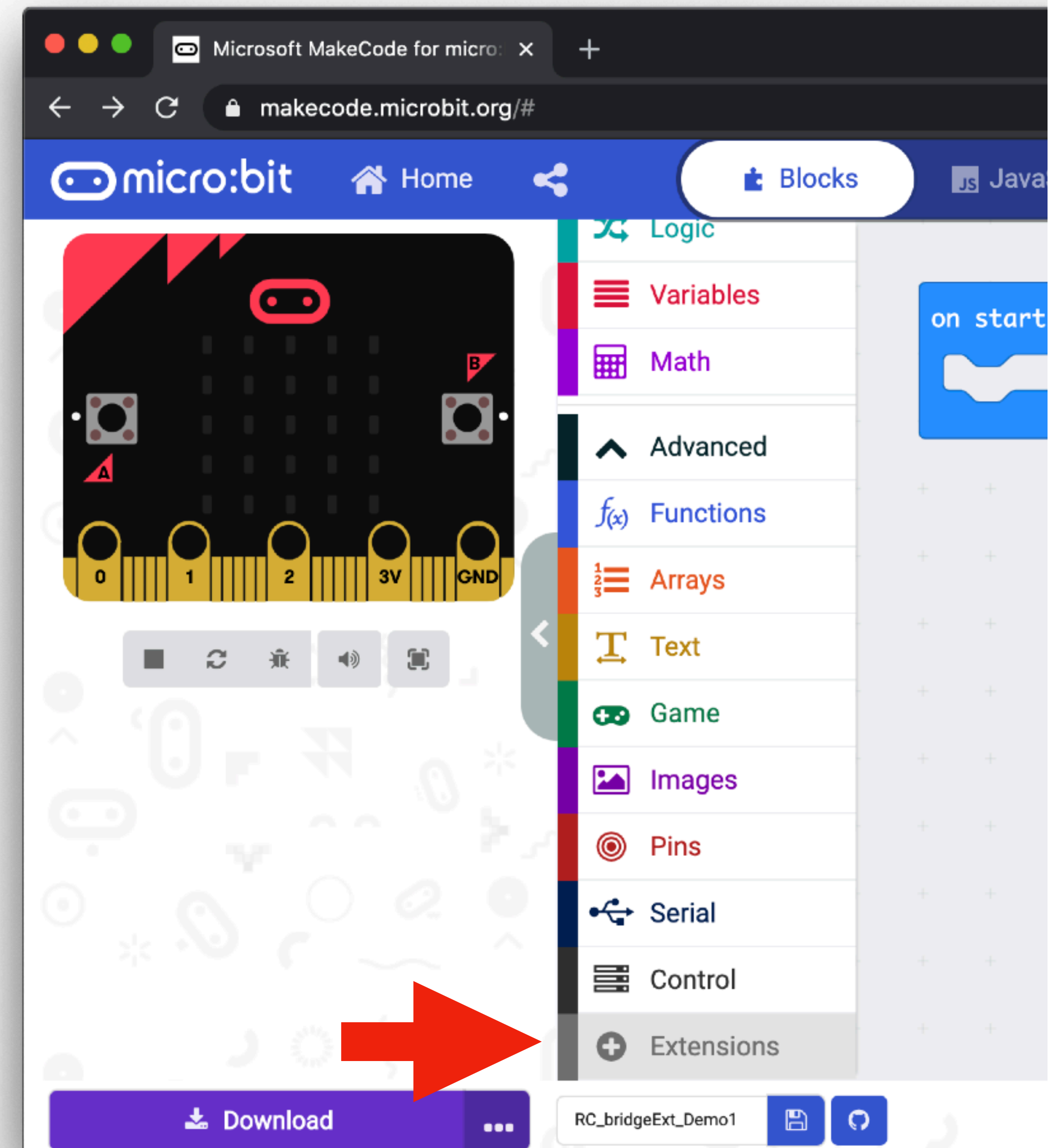


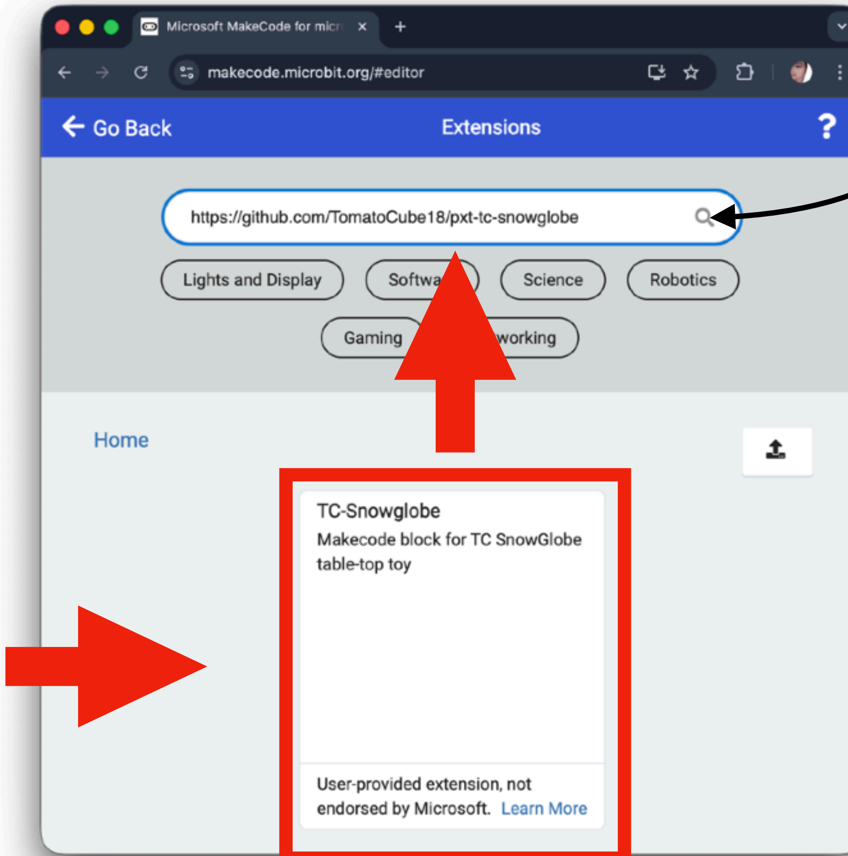


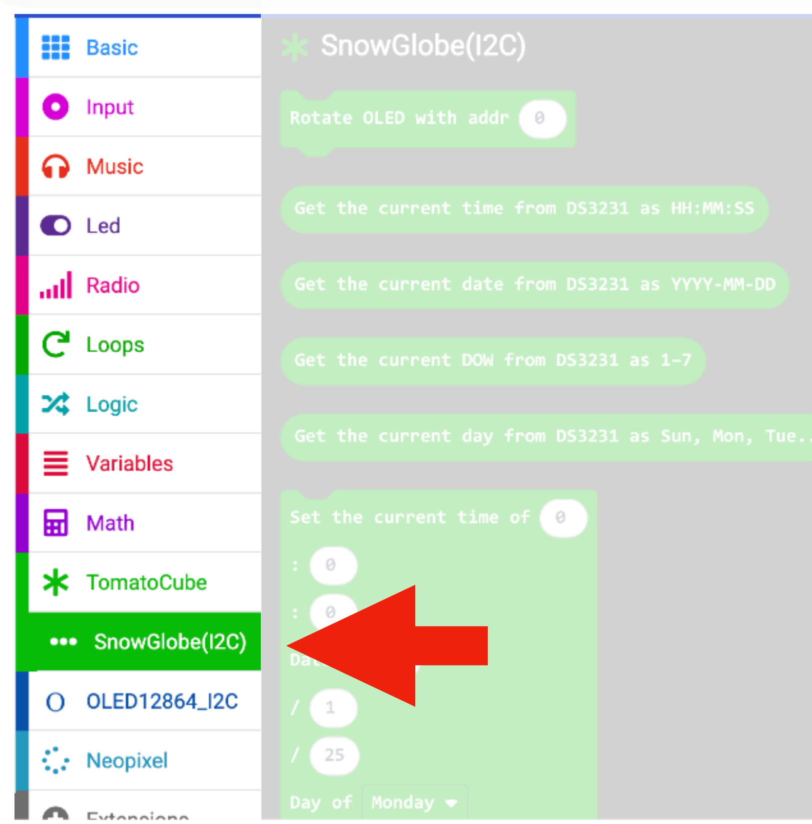


https://github.com/TomatoCube18/pxt-tc-snowglobe

Adding SnowGlobe's MakeCode extension







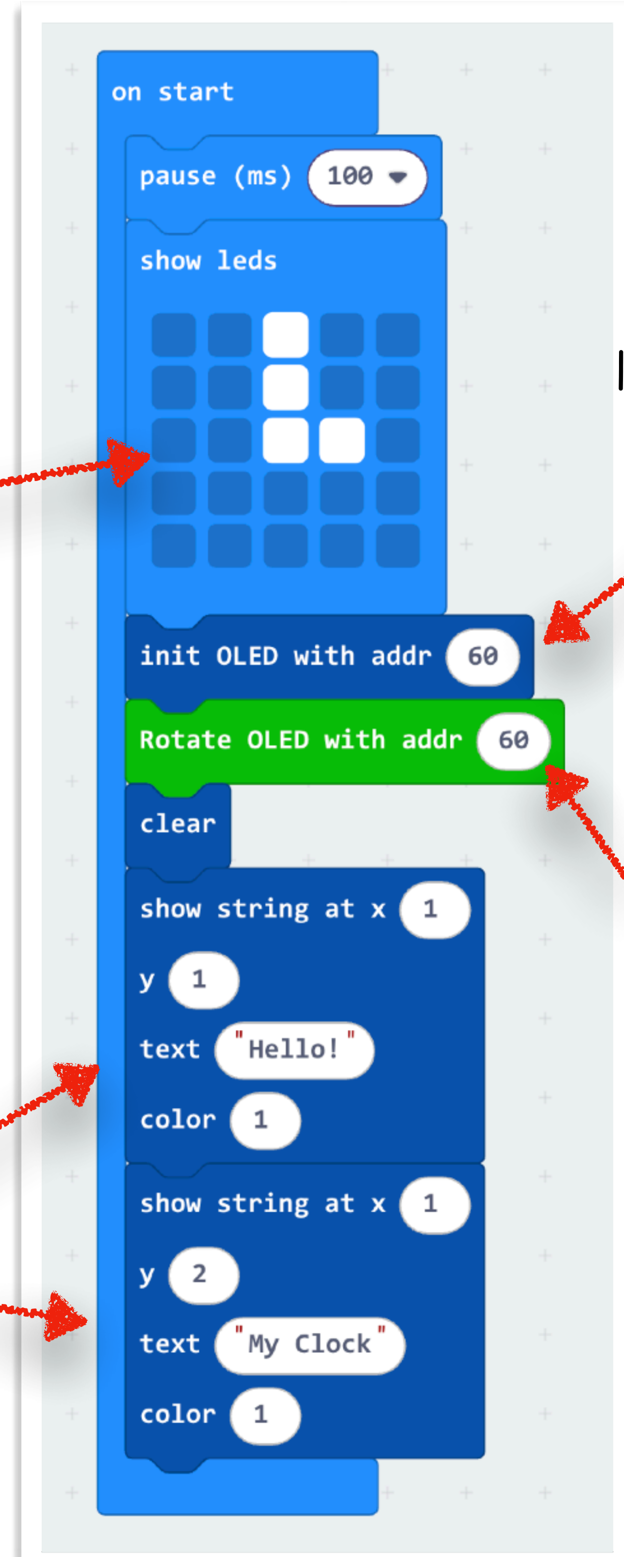
Test Driving the two available Display

Why Addr is 60?

The OLED uses I²C, a communication method where many devices share the same two wires (SDA and SCL). To avoid confusion, each device on the bus has its own unique address, like house numbers on a street. The OLED modules are factory-set to use address 60(0X3C), so when the micro:bit sends commands to address 60, only the OLED listens and responds. This allows the OLED, RTC, and other modules to work together on the same I²C lines without interfering with one another.

Show a crude version of a clock pointing at 3:00 on the LED Display

Just send in the text to be displayed on the screen.

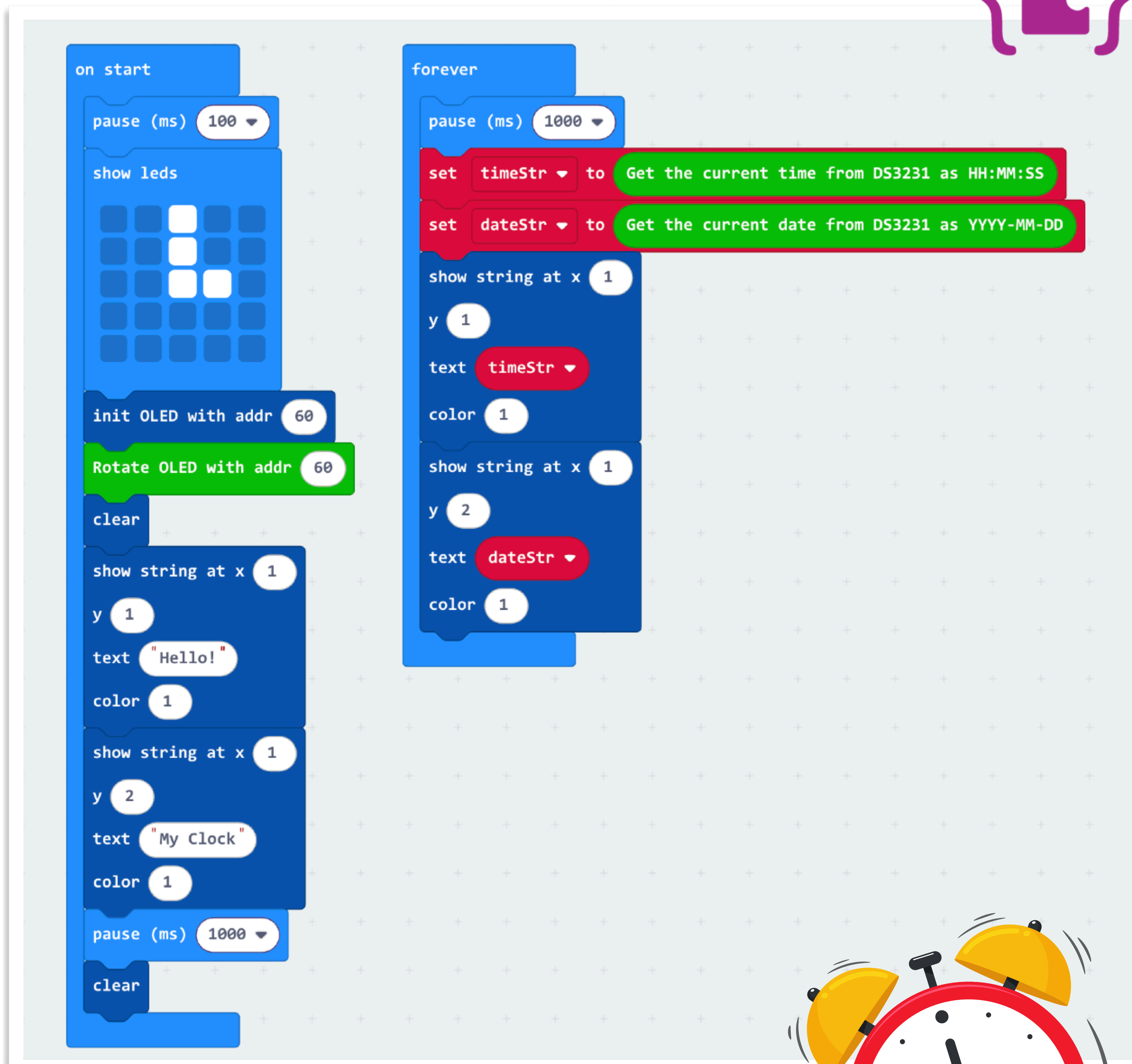


Initialize the extension for the OLED screen.

Rotate OLED screen content by 180°

Building Our SnowGlobe Desk Clock

- By extending the display test code and adding the RTC (Real Time Clock) blocks, we can now bring all the hardware together.
- With just a few extra lines of code, the SnowGlobe becomes a fully working desk clock, showing the real time from the RTC and displaying it clearly on the OLED screen.
- This project demonstrates how multiple I²C devices (OLED + RTC) can work on the same micro:bit, and how combining simple building blocks can transform & create a complete, practical product.



Almost There..

- Part 4 -

Project 1: SnowGlobe Ambient & automatic Nite-light demonstration

- Coming Soon -

Elves are still busy @ work!